Nederlandse norm

NEN-EN 1005-5

(en)

Safety of machinery - Human physical performance - Part 5: Risk assessment for repetitive handling at high frequency

Vervangt NEN-EN 1005-5:2005 2e Ontw.

ICS 13.110-30; 13.180

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Nederlands voorwoord

Voor de in deze norm vermelde normatieve verwijzingen bestaan in Nederland de volgende equivalenten:

vermelde norm	Nederlandse norm	titel
EN 614-1	NEN-EN 614-1	Veiligheid van machines - Ergonomische ontwerpprincipes - Deel 1: Terminologie en algemene principes (en)
EN 614-2	NEN-EN 614-2	Veiligheid van machines - Ergonomische ontwerpprincipes - Deel 2: Interactie tussen het ontwerp van machines en werktaken (en)
EN 1005-2	NEN-EN 1005-2	Veiligheid van machines - Menselijke fysieke belasting - Deel 2: Handmatig hanteren van machines en machine-onderdelen (en)
EN 1005-3:2002	NEN-EN 1005-3:2002	Veiligheid van machines - Menselijke fysieke belasting - Deel 3: Aanbevolen maximale krachten bij machinewerkzaamheden (en)
EN 1005-4:2005	NEN-EN 1005-4:2005	Veiligheid van machines - Menselijke fysieke belasting - Deel 4: Evaluatie van werkhoudingen en bewegingen bij machinewerkzaamheden (en)
EN 1050	NEN-EN 1050	Veiligheid van machines - Principes voor de risicobeoordeling (en,nl)
EN ISO 12100-1	NEN-EN-ISO 12100-1	Veiligheid van machines - Basisbegrippen, algemene ontwerpbeginselen - Deel 1: Basisterminologie, methodologie (en)
EN ISO 12100-2	NEN-EN-ISO 12100-2	Veiligheid van machines - Basisbegrippen, algemene ontwerpbeginselen - Deel 2: Technische beginselen (en,nl)
EN ISO 14738:2002	NEN-EN-ISO 14738:2002	Veiligheid van machines - Antropometrische eisen voor het ontwerp van werkplekken bij machines (en)
ISO/IEC Guide 51	NPR-ISO/IEC Guide 51	Veiligheidsaspecten - Richtlijnen voor hun insluitsel in normen (en)

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EUROPEAN STANDARD

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Safety of machinery - Human physical performance - Part 5: Risk assessment for repetitive handling at high frequency

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Sicherheit von Maschinen - Menschliche körperliche Leistung - Teil 5: Risikobeurteilung für kurzzyklische Tätigkeiten bei hohen Handhabungsfrequenzen

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Foreword

This document (EN 1005-5:2007) has been prepared by Technical Committee CEN/TC 122 “Ergonomics”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by August 2007, and conflicting national standards shall be withdrawn at the latest by August 2007.

As a result of the assessment of the CEN consultant the standard will be published a non-harmonized standard (no reference to Machinery directive and no publication in the Official Journal of EC).

EN 1005 consists of the following Parts, under the general title *Safety of machinery — Human physical performance*:

— *Part 1: Terms and definitions (harmonized standard);*

— *Part 2: Manual handling of machinery and component parts of machinery (harmonized standard);*

— *Part 3: Recommended force limits for machinery operation (harmonized standard);*

— *Part 4 : Evaluation of working postures and movements in relation to machinery (harmonized standard);*

— *Part 5: Risk assessment for repetitive handling at high frequency (non-harmonized standard).*

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

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Introduction

Within the life cycle of a machine from construction to dismantling, various machine-related actions may require repetitive handling at high frequency. Repetitive handling at high frequency can cause musculoskeletal strain and the risk of fatigue, discomfort and musculoskeletal disorders. The designer of a machine should seek to minimise these health risks by taking into account a variety of risk factors including the frequency of actions, the force, postures, durations, lack of recovery and other additional factors.

NOTE 1 Although factors such as duration and lack of recovery periods are relevant factors when assessing risk in relation to human physical performance in the workplace, these factors are controlled by the member states own national legislation, contract agreements with social partners and are not in the scope of this European Standard.

The risk assessment method in this European Standard gives guidance to the designer how to reduce health risks for the operator.

This European Standard is written in conformity with EN ISO 12100-1 and provides the user with guidance for hazard identification for harm through musculoskeletal overload and tools for qualitative and, to an extent, a quantitative risk assessment. The risk assessment tools also indicate how to achieve risk reduction. This European Standard does not deal with risks related to accidents.

The recommendations provided by this European Standard are based on available scientific evidence concerning the physiology and epidemiology of manual work. The knowledge is, however, limited and the suggested guidelines are subject to changes according to future research.

This European Standard is a type B standard as stated in EN ISO 12100-1.

The provisions of this European Standard can be supplemented or modified by a type C standard.

NOTE 2 For machines which are covered by the scope of a type C standard and which have been designed and built according to the provisions of that standard, the provisions of that type C standard take precedence over the provisions of this type B standard.

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1 Scope

This European Standard presents guidance to the designer of machinery or its component parts and the writer of type C standards in assessing and controlling health and safety risks due to machine-related repetitive handling at high frequency.

This European Standard specifies reference data for action frequency of the upper limbs during machinery operation, and it presents a risk assessment method intended for risk reduction option analysis.

This European Standard applies to machinery for professional operation by the healthy adult working population. This European Standard is not applicable for repetitive movements and related risks of the neck, back and lower limbs.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 614-1, *Safety of machinery — Ergonomic design principles — Part 1: Terminology and general principles*

EN 614-2, *Safety of machinery — Ergonomic design principles — Part 2: Interactions between the design of machinery and work tasks*

EN 1005-2, *Safety of machinery — Human physical performance — Part 2: Manual handling of machinery and component parts of machinery*

EN 1005-3:2002*, Safety of machinery — Human physical performance — Part 3: Recommended force limits for machinery operation*

EN 1005-4:2005, *Safety of machinery — Human physical performance — Part 4: Evaluation of working postures and movements in relation to machinery*

EN 1050, *Safety of machinery — Principles for risk assessment*

EN ISO 12100-1, *Safety of machinery — Basic concepts, general principles for design — Part 1: Basic terminology, methodology (ISO 12100-1:2003)*

EN ISO 12100-2, *Safety of machinery — Basic concepts, general principles for design — Part 2: Technical principles (ISO 12100-2:2003)*

EN ISO 14738:2002, *Safety of machinery — Anthropometric requirements for the design of workstations at machinery (ISO 14738:2002)*

ISO/IEC Guide 51, *Safety aspects — Guidelines for their inclusion in standards*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

NOTE Terms and definitions used in EN and ISO standards referred to in this European Standard are also valid for this European Standard.

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3.1
repetitive task
task characterized by repeated work cycles

3.2
work cycle
sequence of technical actions that are repeated always the same way

3.3
cycle time
time elapsing from the moment when one operator begins a work cycle to the moment that the same work cycle is started (in seconds)

3.4
technical action
elementary manual actions required to complete the operations within the work cycle, such as holding, turning, pushing, cutting

3.5
repetitiveness
characteristic of task when a person is continuously repeating the same work cycle, technical actions and movements

3.6
frequency of actions
number of technical actions per minute

3.7
force
physical effort of the operator required to execute the technical actions

3.8
postures and movements
positions and movements of body segment(s) or joint(s) required to execute the technical actions

3.9
recovery time
period of rest following a period of activity in which restoration of human tissue can occur (in minutes)

3.10
additional factors
risk factors which include other factors for which there is evidence of causal or aggravating relationship with work-related musculoskeletal disorders of the upper limb, e.g. vibrations, local pressure, cold environment, cold surfaces

4 Abbreviations

For the purposes of this document, the following abbreviations apply.

Acronyms	Legend for abbreviations
Ad_M	Additional factors Multiplier
ATA	Number of Actual Technical Actions within a shift
CF	“Constant of Frequency” of technical actions per minute
D	net Duration in minutes of each repetitive task
Du_M	Duration Multiplier
FCT	Foreseeable duration of the Cycle Time (in seconds)

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FF	Foreseeable Frequency of technical actions per minute
Fo_M	Force Multiplier
j	generic repetitive tasks
$MSDs$	Musculo-Skeletal Disorders
n	Number of repetitive task/s performed during shift
NEP	Number of Exposed Persons
NPA	Number of Persons Affected by one or more UL-WMSDs
NTC	Number of technical actions in the work cycle
$OCRA$	OCcupational Repetitive Action
PA	Prevalence (%) of persons Affected
RF	Reference Frequency of technical actions per minute
Po_M	Posture Multiplier
Rc_M	Recovery Multiplier
Re_M	Repetitiveness Multiplier
RTA	Number of Reference Technical Actions within a shift
$S.E.$	Standard Error
$UL\text{-}WMSDs$	Upper Limb Work-related Musculo-Skeletal Disorders

5 Requirements

5.1 The application of standards relevant to this document

The designer shall consider the principles given in EN 1050, EN 614-1 and EN 614-2, EN 1005-2, EN 1005-3, EN 1005-4, EN ISO 12100-1 and EN ISO 12100-2 and EN ISO 14738.

5.2 General aspects

The designer of a machine is required to:

a) conduct an assessment of risk of musculoskeletal disorders due to machine related repetitive work;

b) take account in the assessment of the single and combined effects from the most relevant risk factors as repetitiveness, force, working postures, foreseen work duration, lack of recovery periods and additional factors;

c) if possible try to avoid risk ‘at the source’ or alternatively to minimise these health risks by changes in the machinery design (automation, technical aids);

d) when all is done to minimize the risk it is an obligation to inform about residual risks in instructions for use.

Machines and related tasks shall be designed in a way, so that activities demanding high frequency can be performed adequately with respect to the force required, the posture of the limbs and the foreseeable presence of recovery periods. In addition machines and related tasks shall be designed to allow for variations in movements. Additional factors (see 3.11) have to be considered.

When designing machinery and work tasks, the designer shall ensure that the following ergonomics characteristics of well-designed work tasks are fulfilled. These characteristics take into account the differences and dynamic characteristics of the intended operator population, and shall be pursued by designing machinery and machinery related work tasks in interaction (EN 614-2).

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Thus, in design process the designer shall also (see EN 614-2):

i) avoid overload as well as under load of the operator, which may lead to unnecessary or excessive strain, fatigue or to errors. Frequency, duration and intensity of perceptual, cognitive and motor activities shall be designed so as to avoid these consequences;

ii) avoid repetitiveness for the operator, which may lead to unbalanced work strain and thus to physical disorders as well as to sensations of monotony, satiation, boredom or to dissatisfaction.

Short work cycles should therefore be avoided. The operator shall be provided with an appropriate variety of tasks or activities. If repetitive task cannot be avoided:

— cycle time shall not be determined solely on the basis of average time measures or estimated under normal conditions;

— allowances shall be given for deviations from normal conditions;

— very short cycle times shall be avoided;

— opportunities shall be given to the operator to work at his/her own pace, rather than at set pace;

— working on moving objects shall be avoided.

5.3 Risk assessment

5.3.1 General

In this standard risk assessment of musculoskeletal disorders of the upper limb resulting from repetitive handling is described.

The technical action is identified as the specific characteristic relevant to repetitive movements of the upper extremities. The technical action is factored by its relative frequency during a certain time period.

The frequency of technical actions of the upper limbs is related to other risk factors such as force (the greater the force, the lower the frequency), posture (the greater the joint excursion, the longer the time necessary to carry out an action) and recovery periods (if well distributed during the shift, they increase the recovery of muscular function).

Some additional factors can increase the need for force (e.g. awkward tools or personal protective equipment e.g. gloves that interfere with the grasp or movements). Other additional factors can cause damage to human tissue e.g. muscles, tendons and vessels (vibration, compression, cold surfaces).

Data from recent epidemiological studies on workers exposed to repetitive movements of upper limbs allow, among others, designers to forecast from exposure indexes the occurrence of the consequences for Upper Limbs Work-related Musculoskeletal Disorders (UL-WMSDs) [32, 34, 35]. Annex D describes a method of determination. The acceptable situation occurs when the exposure index, given in 5.3.3 (method 2) is not exceeding a level that corresponds to the occurrence of UL-WMSDs as observed in a working population not exposed to occupational risks for the upper limbs [11, 34].

When repetitive handling is unavoidable then a risk assessment and risk reduction approach shall be adopted. In accordance with ISO Guide 51 and EN 1050, this should follow a four-step approach: hazard identification, risk estimation, risk evaluation and risk reduction.

It is recommended to simulate tasks at least once by actual users with a full-size model/prototype of the machinery or the machinery itself (refer to EN 614-1, ergonomic task 'evaluate with users'; see also EN 1005-4:2005, 4.2 ‘Guidance towards risk assessment’).

The following procedure should be adopted when conducting a risk assessment of machinery design involving repetitive handling (Figure 1).

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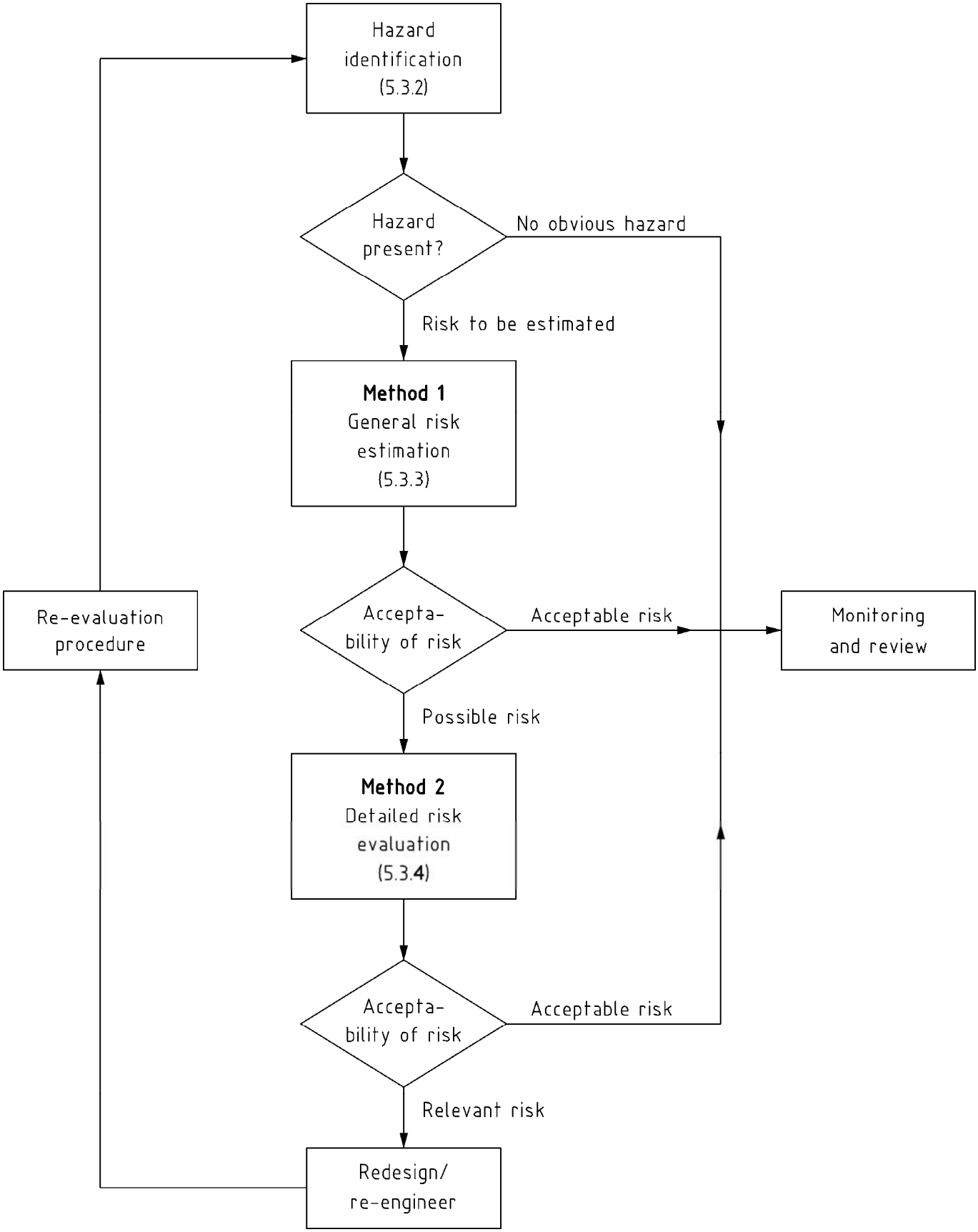

Figure 1 — Risk assessment model

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The first stage of the risk assessment is to identify whether hazards exist which may expose individuals to a risk. If such hazards are present, then a more detailed risk assessment may be necessary. When determining a risk assessment, consideration should be given to the following risk factors:

a) Repetitiveness:
As the movement frequency increases and/or the cycle duration decreases, the risk increases. Frequent repetitive movements giving rise to a risk of musculoskeletal disorders may vary depending on the context of the movement pattern and the individual.

b) Force:
Tasks should involve smooth force exertions, avoiding sudden or jerky movements. Handling precision (accurate picking and placement), the type and nature of the grip may introduce additional muscular effort.

c) Posture and movement:
Work tasks and operations should provide variations to the working posture. The work tasks should avoid extreme ranges of joint movement and there is a need to avoid prolonged static postures. Complex postures involving combined movements (e.g. flexed and twisted) may present greater risk.

d) Duration and insufficient recovery:
Duration can be broken down in a number of ways. The opportunity for recovery or rest can fall within each of these work periods. Insufficient time for the body to recover between repetitive movements (i.e. lack of recovery time) increases the risk of musculoskeletal disorders.

NOTE The designer has no direct influence on the real task duration and recovery time at the machine. He has to refer to a typical scenario of repetitive task duration up to 8 hours per shift with 2 breaks of 10 minutes plus the lunch break. The designer should mention in the "Information for use" if critical values for task duration and recovery time are determined in the risk analysis, e.g. task duration, job duration, and work shift duration.

e) Additional factors:
General consideration should be given to the following additional risk factors:

1) object characteristics (e.g. contact forces, shape, dimensions, coupling, object temperature);

2) vibration and impact forces;

3) environmental conditions (e.g. lighting, climate, noise);

4) individual and organisational factors (e.g. skills, the level of training, age, gender, health problems, pregnancy).

5.3.2 Hazard identification

If the following conditions are satisfied, there is no hazard due to repetitive tasks for upper limbs:

— the task is not characterized by work cycles;

— the task is characterized by work cycles, but perceptual or cognitive activities are clearly prevalent and upper limb movements are residual.

For all the machinery / task combinations in which cyclic manual activities are foreseen, a risk estimation shall be applied.

For each manual task to be performed on machinery, the designer shall:

— identify and count the technical actions (for each upper limb) needed to carry out the work cycle (NTC);

— define the foreseeable duration of the cycle time (FCT);

— consider the force, posture, foreseeable duration and frequency of recovery periods;

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— consider the possibility of rotation between different tasks, at the machinery, e.g. starting procedures, shift of tools and/or settings, loading- and unloading procedures, fetching of materials, maintenance, cleaning'.

A risk estimation model is presented in 5.3.3 (Method 1). A detailed risk evaluation model will be presented in 5.3.4 (Method 2).

When the characteristics described in Method 1 are fully and simultaneously present, it is possible to affirm that risk exposure to repetitive movements at high frequency is acceptable.

Where one or more of the listed characteristics for the different risk factors are not satisfied, the designer shall use a more detailed evaluation (Method 2, 5.3.4).

5.3.3 Risk estimation and simple evaluation of machinery related repetitive handling at high frequency (Method 1)

5.3.3.1 Check of the risk factors

The designer shall check if, considering the main risk factors (force, awkward postures and movements, repetitiveness, frequency of technical actions, additional factors), for each upper limb, the following conditions are satisfied:

a) Absence of force, or use of force in accordance with the criteria regarding the recommended force limits as reported in EN 1005-3.

b) Absence of awkward postures and movements considering the same conditions as summarized below:

   i) the upper arm postures and movements are in the range between 0° and 20° (EN 1005-4:2005, Figure 6, Zone 1);

   ii) the articular movements of the elbow and wrist do not exceed 50 % of the maximum articular range [12, 14], as described in Table 1 and Annex B;

   iii) the kinds of grasp are "power grip", or "pinch lasting no more than 1/3 of the cycle time", as described in Table 1 and Annex B [12, 15, 26].

c) Low repetitiveness.

   This is true if [40, 41]:

   i) the cycle time is more than 30 s.

   ii) the same kinds of technical action are not repeated for more than 50 % of the cycle time.

d) Frequency of technical actions for both upper limbs is less than 40 technical actions per minute. If the frequency is higher than 40 actions per minute for at least one upper limb then move on to Method 2. In order to compute the frequency of technical actions/min (see Annex A for identification of technical action), use the following equation:

$$FF = \frac{NTC \cdot 60}{FCT} \qquad \textbf{(1)}$$

where

FF is the foreseeable frequency of technical actions per minute;

FCT is foreseeable duration of the cycle time in seconds;

NTC is the number of technical actions (for each upper limb) in the work cycle needed to carry out the task.

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e) Absence of additional factors (physical and mechanical factors).

The task should not include hand/arm vibration, shock (such as hammering), localized compression on anatomical structures due to tools, exposure to cold, use of inadequate gloves for grasping etc. [11, 12, 13] (Annex G).

5.3.3.2 Final estimation and evaluation of machinery design by Method 1

When every condition shown in 5.3.3.1, points a), b), c), d) and e) are satisfied for each upper limb, the exposure is acceptable. If one or more conditions mentioned in Method 1 are not met, the designer shall analyse in more detail each risk factor that interferes with the frequency of actions by Method 2.

5.3.4 Detailed risk evaluation of machinery related repetitive handling at high frequency: risk reduction and risk reduction option analysis (Method 2)

5.3.4.1 Evaluation of acceptable frequency of actions when one or more risk factors are present

5.3.4.1.1 General

If one or more conditions defined in Method 1 are not satisfied, the designer shall analyse in more detail each risk factor that has an impact upon the frequency of technical actions. Since different risk factors can be present to a greater or lesser extent, and in a variety of combinations, then different levels of risk can be expected.

The level of risk is assessed with reference to the $OCRA$ method [11, 13, 33]. The $OCRA$ index, when assessing a single repetitive task in a shift (mono task job), is given by the ratio between the foreseeable frequency (FF) of technical actions needed to carry out the task, and the reference frequency (RF) of technical actions, for each upper limb (see Annex A for identification of technical action). This is a particular procedure for mono task jobs. For multitask jobs see Annex H.

In this context:

$$OCRA\ index = \frac{FF}{RF}$$

The foreseeable frequency (number per minute) of technical actions needed to carry out the task (FF) is given by the following equation:

$$FF = \frac{NTC \cdot 60}{FCT}$$

where

FCT is foreseeable duration of the cycle time in seconds;

NTC is the number of technical actions (for each upper limb) needed to carry out the task during one cycle.

The following equation calculates the reference frequency (number per minute) of technical actions (RF) on a work cycle base:

$$RF = CF \times Po_M \times Re_M \times Ad_M \times Fo_M \times (Rc_M \times Du_M)$$

where

CF is the “constant of frequency” of technical actions per minute = 30;

Po_M; Re_M; Ad_M; Fo_M are multipliers for the risk factors postures, repetitiveness, additional, force;

Rc_M is the multiplier for the risk factor “lack of recovery period”;

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Du_M is the multiplier for the overall duration of repetitive task(s) during a shift.

When designing a machinery related task, evaluate reference frequency of the technical actions within a work cycle that is representative of the task under examination. The analyses shall include the main risk factors that the designer can influence with the consequent choice of a specific multiplier for each risk factor. These multipliers will decrease from 1 to 0 as the risk level increases. The risk factors and the corresponding multiplier, influenced by the designer, are:

— awkward or uncomfortable postures or movements (posture multiplier) (Po_M), see 5.3.4.1.2;

— high repetition of the same movements (repetitiveness multiplier) (Re_M), see 5.3.4.1.3;

— presence of additional factors (additional multiplier) (Ad_M), see 5.3.4.1.4;

— frequent or high force exertions (force multiplier) (Fo_M), see 5.3.4.1.5.

The other factors considered in the equation ($Rc_M \times Du_M$) are generally out of the direct influence of the designer and consequently they will be considered, in this context, as a constant, reflecting a common condition of repetitive task duration of 240 min to 480 minutes/shift with 2 breaks of 10 min plus the lunch break.

In practice, to determine the reference frequency (per minute) of technical actions (RF), proceed as follows:

— start from CF (30 actions per minute);

— CF (the frequency constant) has to be weighted (by the respective multipliers) considering the presence and degree of the following risk factors: force (Fo_M), posture (Po_M), repetitiveness (Re_M) and additional (Ad_M);

— apply the constant that considers the multiplier for repetitive task duration (Du_M) and the multiplier for recovery periods (Rc_M);

Thus the value obtained represents the reference frequency (per minute) of technical actions (RF) for the examined task in the common condition of at least 2 breaks of 10 min (plus the lunch break) in a shift of maximal 480 min.

5.3.4.1.2 Posture Multiplier (Po_M)

If the conditions for posture are the same as in Method 1, the multiplier is 1. If those conditions are not present, use the indications in Table 1 for obtaining the specific multiplier factor (Po_M) (see also Annex B):

When considering awkward postures and movements of operators in Table 1, it is important to determine the range of body dimensions of the user population and the general design principles, which are described in EN 614-1.

Early in the design process a comparison shall be made between the body dimensions of the user population and the machinery dimensions. This can be done by means of standards (EN 547-1, EN 547-2, EN 547-3 and EN ISO 14738), body templates or computer manikins.

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Table 1 — Multipliers for awkward postures

Awkward posture [12] See Figures B.1 and B.2	Portion of the cycle time			
	Less than 1/3 from 1 % to 24 %	1/3 from 25 % to 50 %	2/3 from 51 % to 80 %	3/3 more than 80 %
Elbow supination (≥ 60°)	1	0,7	0,6	0,5
Wrist extension (≥ 45°) or flexion (≥ 45°)				
Hand pinch or hook grip or palmar grip (wide span)				
Elbow pronation(≥ 60°) or flexion/extension (≥ 60°)	1	1	0,7	0,6
Wrist radio-ulnar deviation (≥ 20°)				
Hand power grip with narrow span (≤ 2 cm)				

At the end of the analysis of awkward postures, select the lowest multiplier Po_M in accordance with the foreseen postures and movements of elbow, wrist and hand (type of grip) for calculating the equation.

The designer, at this step, shall consider also shoulder postures and movements (for details see EN 1005-4, EN ISO 14738).

NOTE Any movement above shoulder height should be avoided. At this moment there is no available data for identifying a specific Po_M for shoulders: consequently Po_M for shoulders cannot be included in the $OCRA$ index computation procedure.

Annex B gives further explanation how to analyse postures and movements of the upper limbs.

5.3.4.1.3 Repetitiveness multiplier (Re_M)

When the task requires the performance of the same technical actions of the upper limbs for at least 50 % of the cycle time or when the cycle time is shorter than 15 s, the corresponding multiplier factor (Re_M) is 0,7. Otherwise Re_M is equal to 1.

5.3.4.1.4 Additional factors multiplier (Ad_M)

The main additional factors are (non exhaustive list): use of vibrating tools, gestures implying counter shock (such as hammering), requirement for absolute accuracy, localized compression of anatomical structures, exposure to cold, use of gloves interfering with handling ability, high pace completely determined by the machinery.

If additional factors as listed above are absent for most of the task duration, the additional factor multiplier (Ad_M) equals 1. Otherwise the additional factor multiplier (Ad_M) equals:

— Ad_M = 1, if additional factor(s) are present at the same time for less than 25 % of the cycle time;

— Ad_M = 0,95, if additional factor(s) are present at the same time for 1/3 (from 25 % to 50 %) of the cycle time;

— Ad_M = 0,90, if additional factor(s) are present at the same time for 2/3 (from 51 % to 80 %) of the cycle time;

— Ad_M = 0,80, if additional factor(s) are present at the same time for 3/3 (more than 80 %) of the cycle time.

Annex G further explains how to identify and evaluate the different additional factors.

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5.3.4.1.5 Force Multiplier (Fo_M)

If the criteria described in Method 1 are satisfied, the multiplier is 1.

If these conditions are not met, use Table 2 to determine the force multiplier (Fo_M) that applies to the average level of force, as a function of time.

The force level (upper row) is given as a percentage of the Maximal Isometric Force (F_b) as determined in EN 1005-3:2002, 4.2.1.

When using a mock-up for risk assessment, a value derived from the application of the CR-10 Borg-scale can be used (second row) [8, 9, 15, 18, 36].

Use a Fo_M = 0,01 when the technical actions require 'peaks' above 50 % of F_b or a score of 5 (or more) in CR-10 Borg scale for almost 10 % of the cycle time.

The values in the Table 2 can be interpolated if intermediate results are obtained.

Annex C further explains how to determine the force level.

Table 2 — Multiplier relative to the different use of force (Fo_M)

Force level in % of F_b	5	10	20	30	40	≥ 50
CR-10 Borg	0,5	1	2	3	4	≥ 5
Score	very, very weak	very weak	weak	moderate	somewhat strong	strong/very strong
Force multiplier (Fo_M)	1	0,85	0,65	0,35	0,2	0,01

5.3.4.1.6 Predetermined value (Constant) for the repetitive task duration multiplier (Du_M) and the multiplier for recovery periods (Rc_M)

Since the multipliers (Du_M and Rc_M), considered in the equation, are generally out of the direct influence of the designer, they are here considered as unique constant, reflecting a common condition as:

Du_M = 1 (multiplier for overall repetitive task duration of 240 min to 480 min);

Rc_M = 0,6 (for a foreseeable presence of 2 breaks of 10 minutes and a lunch break in a repetitive task duration of 240 min to 480 min per shift).

Therefore: ($Rc_M \times Du_M$) = 0,6

If the risk analysis of the machine shows that the risk is not acceptable when the total task duration exceeds a specific value or the recovery time is shorter than a specific value, this shall be mentioned in the "information for use", see Annex E.

5.3.4.1.7 Risk reduction analysis

When the result of the risk evaluation shows an unacceptable risk level the designer should reduce it by optimising one or more of the following factors:

- number of technical actions needed in a work cycle;
- cycle time;

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- awkward postures;
- level of force of the technical actions;
- additional factors.

He can also inform the user that it is possible to reduce the risk by reducing the task duration, adding breaks or introducing job rotation.

Regarding options for risk reduction see the example of Annex F.

5.3.4.2 Final evaluation by Method 2

The *OCRA* Index is obtained (for jobs with a single repetitive task) by comparing, for each upper limb, the foreseeable frequency (*FF*) of technical actions needed to carry out the repetitive task and the reference frequency (*RF*) of technical actions by means of this equation:

$$OCRA\ index = \frac{FF}{RF}$$

Table 3 supplies the relevant values of the *OCRA* index to assess the risk in relation to the 3-zone rating system (green, yellow, red) and to decide for consequent actions to be taken.

Table 3 — Classification of *OCRA* Index results for evaluation purposes

OCRA Index	Zone	Risk evaluation
≤ 2,2	Green	Acceptable
2,3 to 3,5	Yellow	Conditionally acceptable
> 3,5	Red	Not acceptable

When a 'conditionally acceptable' condition occurs, the designer shall:

— reconsider the design of the machinery and of the task in order to obtain an acceptable condition;

— refer to Annex E which gives the criteria about recovery periods and daily duration of the task as a basis or details in the information for use.

Annex D reports information about the criteria adopted for *OCRA* index classification and about forecasting models of the expected percentage of Persons Affected (*PA*) by one or more Upper Limb Work Related Musculo-Skeletal Disorders (UL-WMSDs) at a given *OCRA* index value.

Annex F provides a full example for the use of Method 2.

6 Verification

For each safety requirement and/or protective measure, except if it is self-evident, a method of verification shall be established:

- a) by testing (e.g. functional test of a new sequence of technical actions);
- b) by measurement (e.g. measurement of required forces and postures versus time);
- c) by calculation (e.g. recalculation of *OCRA* Index after improvements);
- d) by any other method of verification, if testing and calculation are not adequate (e.g. by visual inspection).

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7 Information for use

The designer is compelled by the machinery directive to inform the users about any health risks that can be foreseen due to manual repetitive tasks related to the machine. Therefore the designer should prepare “Information for use” that describes the residual risk that can not be avoided after all has been done to achieve an acceptable risk.

The "information for use" of a machine should give instructions to ensure safe and proper use of the machine. It should also warn the user for any residual risks related to the use of the machine if these instructions are not followed (EN ISO 12100-2 and CEN Guide 414). All conditions that are specified in the risk assessment and could be influenced by the user shall be mentioned in these specifications. It should also indicate whether specific training is needed with respect to a proper use of the machine.

The ‘Information for use’ should also contain information on possible risk factors that can be foreseen but are out of the designer’s scope, i.e.:

1) The duration and recovery time of the user of the machine.
   Usually the designer of machinery has no influence on the actual task duration and recovery time of the users of the machine. Therefore, in methods 1 and 2 of this standard a usual organizational context is assumed in which repetitive task duration lasts 240 min to 480 min per shift and two (plus the lunch break) breaks of 10 min (per shift) are provided. The risk of alternative task durations and recovery times are described in Annex E. The designer can use this information for the ‘information for use’. He should also mention possible solutions for risk reduction, such as reducing task duration, providing more breaks and introducing job rotation. An example of the effect of adding breaks is given in Table F.17 of Annex F. An example of introducing job rotation consisting in different repetitive tasks, with a consequent risk evaluation by Method 2 is given in Annex H.

2) Additional risk factors.
   Apart from the main risk factors that are dealt with in Method 2, there could be additional factors present that may increase the risk. Examples of these additional risk factors are: use of vibrating tools; exposure to cold or precise positioning of a piece or an object. Annex G describes these and other additional factors. These additional factors should be taken into consideration by the user of the machine and should therefore be mentioned in the ‘Information for use’.

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Annex A
(informative)

Identification of technical action

A.1 General

Technical actions imply musculoskeletal activity of the upper limbs. They should not be identified with a single joint movement, but rather with a complex movement involving one or more joints and segments to enable the completion of a simple working task [12, 13]. The task analysis methods, generally used in industry, identify the elementary movements of a given operation to determine the time required to accomplish the operation.

NOTE The task analysis methods generally used in industry identify the elementary movements of a given operation to determine the time required to accomplish the operation. The most common methods are [6, 7, 10, 16, 17, 20, 21, 22, 23, 24, 25, 27, 28, 31, 37, 38, 39, 42, 44, 45].

- Chronometer analysis

- Predetermined Time Systems (PTS) like: MTA (Motion Tyme Analysis), MTS (Motion Time System), WF (Work Factor), MTM 1 (Motion Time Measurements), MTM 2, MTM 3, MTM V, MTM MEK (Motion Time Measurements), MEK, MTM UAS (Motion Time Measurements – Universelle Analysier System), MODA PTS (Modular Analysis Predetermined Time Systems).

The technical actions are more similar (even if not identical) with the “elements” considered in the task analysis methods listed in the note. So the technical actions are more easily recognisable by technicians because their identification and the task analysis methods both aim towards the description of the technical movements carried out by the operator to complete a work cycle. Table A.1 lists the criteria for the definition and counting of technical actions.

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Table A.1 — Criteria for the definition and counting of technical actions

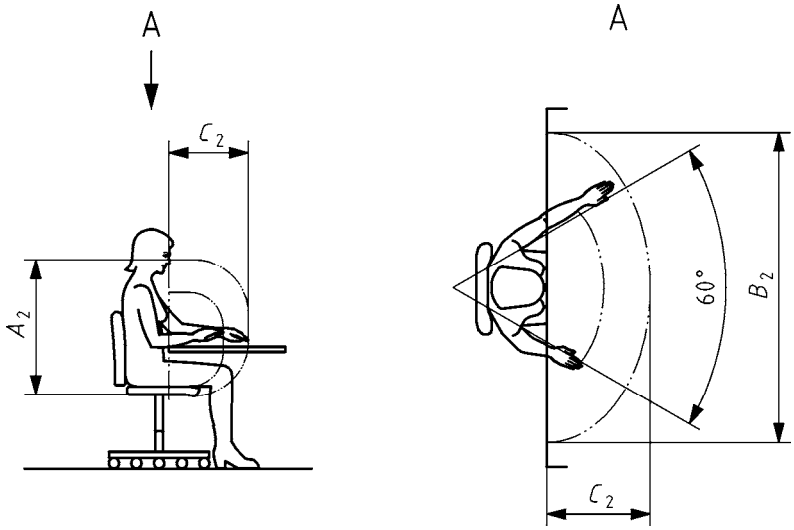
Technical action	Criteria for the definition and counting
Move	Means transporting an object to a given destination by using the upper limbs (without walking). Moving an object should be considered as an action exclusively when: - the object weighs more then 2 kg (with the hand in grip) or 1 kg (with the hand in pinch) (see Annex B) and - the upper limb has a wide movement covering a distance of > 1 m.
Reach	Means shifting the hand towards a pre-fixed destination Reaching an object should be considered as an action exclusively when the object is positioned beyond the reach of the working area limits (A2, B2, C2) as given by EN ISO 14738: 2002  Maximum working area height (A2): 730 mm Maximum working area width (B2): 1 170 mm Maximum working area depth (C2): 415 mm

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Table A.1 (*continued*)

Technical action	Criteria for the definition and counting
Grasp/take	Gripping an object with the hand or fingers, to carry out an activity or task, is a technical action. Synonyms: take, hold, grip again, take again.
Grasp with one hand, Grasp again with other hand	The action of passing an object from hand to hand is considered as two separate actions: one for the right hand (Grasp with one hand) and one for the left (Grasp with other hand)
Position	Positioning an object or a tool in a pre-established point constitutes a technical action. Synonyms: position, lean, put, arrange, put down; equally, re-position, put back etc.
Putting in, Pulling out	The action of putting in or pulling out needs to be considered as a technical action, when the use of force is required. Synonyms: to insert, to extract.
Push/pull	These need to be counted as actions because they stem from the need to apply force, although maybe only a little, in order to obtain a specific result. Synonyms: to tear, to press.
Release, let go	If, once an object is no longer necessary, it is simply “released” by opening the hand, or the fingers, then the action need not be considered as a technical action.
Start-up	This needs to be considered as an action when start-up of a tool requires the use of a push-button or lever by parts of the hand, or by one or more fingers. If start-up is done repeatedly, then consider one action for every start-up. Synonyms: press button, lift/lower lever.
Specific actions during a phase	In addition to those listed here, many technical actions exist that specifically describe the processing of a part/object, i.e.: to bend or fold to bend or curve, deflect to squeeze, rotate, turn to settle, to shape to lower, hit, beat to brush (count each brush passage on part to be painted) to grate (count each passage on part to be grated) to smooth or polish (count each passage on part to polish) to clean (count each passage on part to clean) to hammer (count each single hit on part) to throw etc. Each one of these actions need to be described and counted once for every repetition, i.e.: — turn twice = 2 technical actions; — lower 3 times = 3 technical actions; — 4 brush strokes = 4 technical actions.
Walk, visual inspection	These need not be considered as technical actions because they do not imply any activity of the upper limbs.
Carry	Means walking, carrying an object to a give destination. Carrying an object should be considered as an action exclusively when: — the object weights more then 2 kg (with the hand in grip) or 1 kg (with the hand in pinch), see Annex B and — the upper limb has a wide movement covering a distance of > 1 m.
NOTE Identical actions need still be counted every time that they are repeated. It needs to be remembered that, for defining the frequency of action (number of actions per minute), the single technical actions are counted and not their duration.	

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A.2 Examples for identifying and counting technical actions

A.2.1 Example 1: Pick and place (Tables A.2 and A.3)

The operation, described here, concerns the picking up of an object (a cylinder) from a container, and then placing it into a hole on the workbench which is close to the body. Basically, this is what we call PICK (first technical action) and PLACE (second technical action) an object being processed. In this example only the right upper limb is working: 2 technical actions are present in this cycle only for the right upper limb.

After identifying the technical actions, count them in the cycle and, timing the cycle length in seconds, use the following equation (for each upper limb) to calculate the frequency of technical actions:

Number of actions in the cycle (for right and left limb separately) x *60/cycle time*

If it is necessary to re-grasp and re-position the work piece, the "re-grasping" and re-positioning would count as two new actions (Table A.3).

Table A.2 — The OCRA method for counting technical actions of the pick and place (Example 1)

	Technical actions	
	Left upper limb	Right upper limb
		1 pick up cylinder
		1 place cylinder in hole
Total number	0	2
Cycle time duration (in seconds)	6	6
Frequency (Number of technical actions per minute)		20

Table A.3 — The OCRA method for counting technical actions of the pick and place task, re-grasp and re-position (Example 1)

	OCRA technical actions	
	Left upper limb	Right upper limb
		1 take cylinder
		1 place cylinder in hole
		1 re-grasp
		1 re-position
Total number	0	4
Cycle time duration (in seconds)	6	6
Frequency (Number of technical actions per minute)		40

A.2.2 Example 2: Pick and place with transfer from one hand to the other and with visual inspection (Table A.4)

The operation described here is a pick and place operation with transfer of the piece from one hand to the other and visual inspection.

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The operator grips the cylinder with the left hand, passes it to the right hand, rotates it for a visual inspection and, still with the right hand, positions it in the required place. When counting technical actions, 'visual inspection' is not considered, because it does not require any mechanical action of the upper limbs. In this case, the operator actually rotates the cylinder to inspect it visually: therefore, this control is a mechanical action, and needs therefore to be counted as a technical action (rotation).

Table A.4 — The OCRA method for counting technical actions of the pick and place task with transfer from left to right hand and visual inspection (Example 2)

	Technical actions	
	Left upper limb	Right upper limb
	1 take cylinder	
		1 grasp cylinder
		1 rotate cylinder
		1 position cylinder
Total number	1	3
Cycle time duration (in seconds)	6	6
Frequency (number of technical actions per minute)	10	30

A.2.3 Example 3: Pick and place while transporting a load (Table A.5)

In this case, the operator needs to transport a load weighing 4 kg, from a container which is over 1 m distance from the workbench, to the workbench itself. The technical actions that have been listed are: grip the part, transport the load and place it.

TRANSPORT has to be counted as a technical action of the upper limb(s) only when the load weighs more then 2 kg per arm (with the fingers in grip) or 1 kg per arm (with the fingers in pinch) (see Annex B), and it is transported for at least 1 m (two steps).

Table A.5 — The OCRA method for counting technical actions of the pick and place with transport (Example 3)

	Technical actions	
	Left upper limb	Right upper limb
		1 grasp load
		1 transport load with arm
		1 position load on bench
Total number	0	3
Cycle time duration (in seconds)	6	6
Frequency (Number of technical actions per minute)	0	30

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A.2.4 Example 4: Cyclical use of a tool with repeated and identical actions (Table A.6)

The operator, using a drill, needs to make 3 holes at 3 different points. After gripping the drill with his right hand (1st action), he places it over the point where the hole needs to be drilled, pushes the button to start the drill, pushes the drill to obtain the hole, and extracts the drill. These 4 actions are repeated 3 times (total 12 technical actions). At the end the drill is put down. The total number of technical actions is therefore 14, all carried out by the right upper limb.

OPERATE describes the action of using the hand or finger/s to operate the drill; PUSH indicates the need to apply force, even if minimal; REMOVE indicates the need to perform the operation using force even if minimal; PLACE describes the need to place the tool in a predetermined spot. If the tool were suspended and returned to its original position passively, the RELEASE action would not be counted.

Table A.6 — The OCRA method for counting technical actions during the cyclical use of a tool with repeated identical actions (Example 4)

	Technical actions	
	Left upper limb	Right upper limb
		1 grasp drill
		1 place on 1st hole
		1 operate by pressing button
		1 push to make 1st hole
		1 remove drill
		1 place on 2nd hole
		1 operate by pressing button
		1 push to make 2nd hole
		1 remove drill
		1 place on 3rd hole
		1 operate by pressing button
		1 push to make 3rd hole
		1 remove drill
		1 replace drill
Total number	0	14
Cycle time duration (in seconds)	14	14
Frequency (Number of technical actions per minute)	0	60

A.2.5 Example 5: Technical actions not carried out in every cycle (Table A.7)

There are cases in which some of the technical actions are not carried out in every cycle, but once every few cycles. These actions need to be counted within each cycle as fractions of actions. In the example re-grasp and re-place have to be done every two cycles: each of them need to be counted as 0,5 technical actions per cycle.

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Table A.7 — The OCRA method for counting technical actions of the pick and place task, re-grasp and re-position (Example 5)

	Technical actions	
	Left upper limb	Right upper limb
		1 take cylinder
		1 place cylinder in hole
		½ re-grasp
		½ re-position
Total number	0	3
Cycle time duration (in seconds)	6	6
Frequency (Number of technical actions per minute)		30

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Annex B
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Posture and types of movements

Upper limb postures and movements, during repetitive tasks, are of fundamental importance in contributing towards the risk of various musculoskeletal disorders. Much agreement can be found in the technical literature as to the potential damage from awkward postures and movements of each joint, from postures maintained for a long time (even if not extreme), and from specific, repetitive movements of the various segments.

The analysis of postures and movements will concentrate on each single segment of the upper limbs (hand, wrist, elbow, shoulder) and is aimed at checking the presence and time pattern in the cycle (frequency, duration) of static postures and dynamic movements involving each of the segments/joints considered.

The description may be more or less analytical but needs at least to address the following items:

a) Technical actions requiring postures or movements of a single segment beyond a critical level of angular excursion.

b) Technical actions involving static postures and/or movements that, even in acceptable angular excursion, are maintained or repeated in the same way (repetitiveness).

c) The duration, expressed as a fraction of cycle/task time, of each of the conditions reported above.

The combination of these descriptive factors (posture/time) will provide the classification of effort for each segment considered.

NOTE In order to identify the so-called angular excursion critical levels (awkward postures and movements), reference can be made to EN 1005-4 and if necessary to data and proposals available in the literature [5, 11, 12, 14, 19, 26, 29, 30, 38] which are quite convergent, though differing in the level of analytical detail (inclusion/exclusion of some kinds of movement; critical excursion values of main movements).

An accurate description of posture and movements can also be considered as a predictive element for specific pathologies of the upper limbs, which can be foreseen for exposed operators in the presence of other risk elements (such as frequency, force, duration etc.)

The description/assessment of the postures and movements needs to be done over a representative cycle for each of the repetitive tasks examined. This needs to be via the description of the duration of the postures and/or movements of the four main anatomical segments (both right and left):

1) posture and movements of the arm with respect to the shoulder (flexion, extension, abduction);

2) movements of the elbow (flexions-extensions, prono-supinations of the forearm);

3) postures and movements of the wrist (flexions-extensions, radio-ulnar deviations);

4) postures and movements of the hand (mainly the types of grip).

To simplify the analysis of postures and movements for the designer, an awkward posture related to a technical action is assessed as present if joint segment travels over an angle greater than 50% of maximum range of motion of this specific joint (or if an awkward position for gripping with the hand is present) (see also Figures B.1, B.2 and B.3).

Awkward postures are classified by different scores extrapolated from the data on the subjective perception of different joint involvement [12].

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When studying the postures and movements of the shoulder, mention is to be made of a recent study [35] that showed an increased risk of shoulder disorders when the arm is moved or maintained at about shoulder level (extreme elevation) for more than 10 % of the cycle time.

As far as the types of handgrip are concerned, some of them (pinch, upper palmar grip, hook grip, narrow span) are considered as being less favourable than the power grip and are therefore classified as implying medium/high involvement.

The figures in this annex refer to the main joint movements (EN 1005-1 and EN 894-3) of the upper limbs (Figures B.1 and B.2) and, for the hand, the different type of grip (Figure B.3): the Table 1 in 5.3.2.1 summarizes the degrees beyond 40 % to 50 % of joint excursion range.

Posture evaluation involves the five operating steps described below:

1) the description of the postures and/or movements, done separately for the right and left joints;

2) establishing if there is joint involvement in a risk area (awkward postures and/or movements), and its timing within the cycle:

   — 1/10 from 10 % to 24 % of the cycle time;

   — 1/3: from 25 % to 50 % of the cycle time;

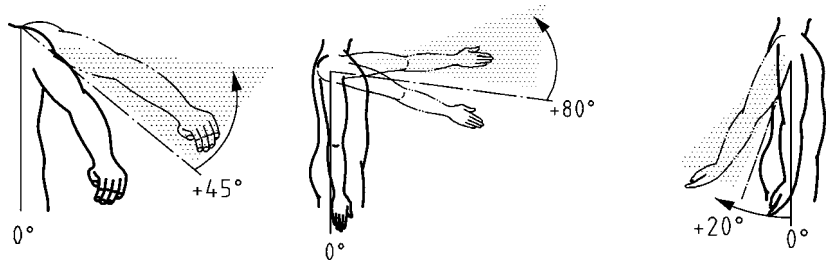
   — 2/3: from 51 % to 80 % of the cycle time;

   — 3/3 more than 80 % of the cycle time;

3) finding (Table B.1) the corresponding Posture multiplier (Po_M);

4) establishing the presence of repetitiveness of certain movements which can be pinpointed by observing technical actions, or groups of technical actions which are all equal to each other for at least 50 % of the cycle time, or by the presence of static positions which are maintained for at least 50 % of the cycle time, or by a very short duration of the cycle (less than 15 s but obviously characterized by the presence of actions of the upper limbs);

5) consider the corresponding Repetitiveness multiplier (Re_M).



a) Lateral elevation abduction/adduction (100 % of joint range is 90º, awkward posture > 45º)

b) Frontal elevation flexion (100 % of joint range is 180º, awkward posture > 80º)

c) Extension (100 % of joint range is 40º, awkward posture > 20º)

Figure B.1 — Shoulder postures and movements

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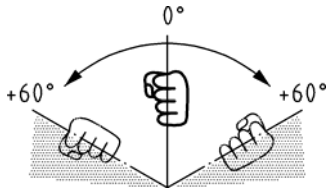
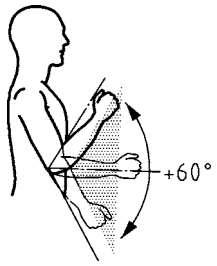
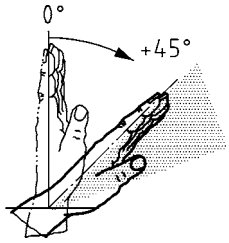


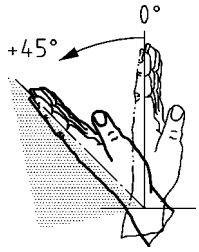
a) Elbow – prono suppination (100 % of joint range is 90º, awkward posture > 60º)



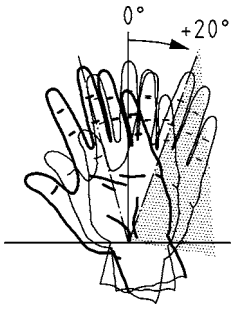
b) Elbow – flexion, extension (100 % of joint range is ± 150º, awkward posture > 60º)



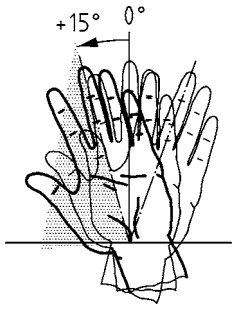
c) Wrist – Palmar flexion (100 % of joint range is 90º, awkward posture > 45º)



d) Wrist – Dorsal extension (100 % joint range is 90º, awkward posture > 45º)



e) Wrist – Ulnar deviation (100 % of joint range is ± 40º, awkward posture > 20º)



f) Wrist – Radial deviation (100 % range is ± 30º, awkward posture > 15º)

Figure B.2 — Elbow and wrist postures and movements

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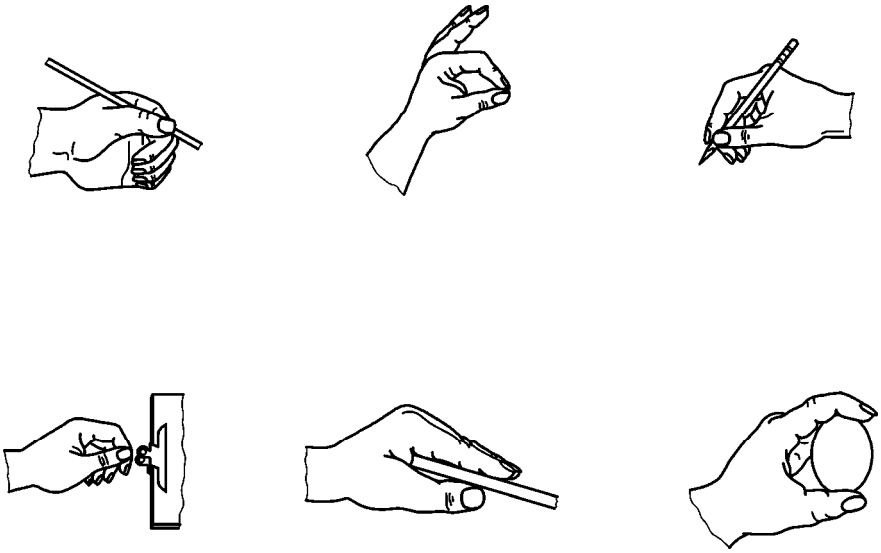
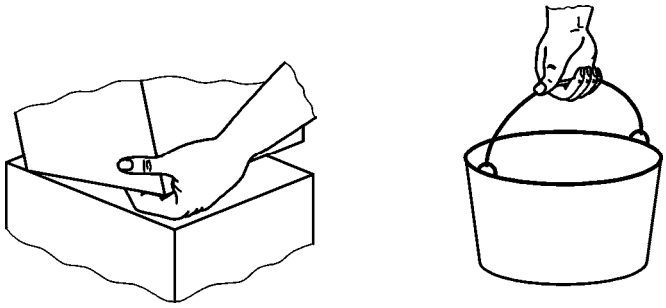
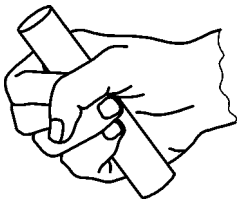


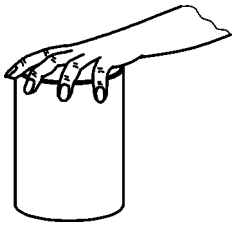
a) 6 examples for pinch



b) 2 examples for hook grips



c) Power grip



d) Palmar grip

Figure B.3 — Different kinds of hand grip

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Table B.1 — Definition of awkward postures and movements and their corresponding Posture multipliers (Po_M)

Awkward posture [12]	Portion of the cycle time			
	Less than 1/3 from 1 % to 24 %	1/3 from 25 % to 50 %	2/3 from 51 % to 80 %	3/3 more than 80 %
Elbow supination (60°)	1	0,7	0,6	0,5
Wrist extension (≥ 45°) or flexion (≥ 45°)				
Hand pinch or hook grip or palmar grip (wide span)				
Elbow pronation(≥ 60°) or flexion/extension (≥ 60°)	1	1	0,7	0,6
Wrist radio-ulnar deviation (≥ 20°)				
Hand power grip with narrow span (≤ 2 cm)				

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Annex C
(informative)

Force

C.1 General

C.1.1 Introduction

Force more directly represents the biomechanical involvement necessary to carry out a given technical action, or sequence of technical actions. Force may be intended as being external, applied force, or internal, tension developed in the muscle, tendon and joint tissues. The need to develop force during work-related technical actions may be related to the moving or the keeping still of tools and objects, or to keep a part of the body in a given position. The use of force may be related to static actions (contractions), or to dynamic actions (contractions). When the first situation occurs, it is generally described as static load, which some authors describe as a distinct risk element [19].

The need for using force repetitively is considered as a risk factor for tendon and muscle disorders. Furthermore, a multiplicative interaction has been shown between force and frequency of technical actions, especially for disorders affecting tendons or nerves.

Force quantification in actual work situations is difficult. Some authors use semi-quantitative estimation of external force via the weight of the objects being handled. In other cases, it has been suggested to use mechanical or electronic dynamometers. Surface electromyography techniques can be used to quantify internal forces exerted by muscles. All of these methods present implementation difficulties.

Effects of physical loads will be estimated by force multipliers Fo. Force multipliers may be determined in two different ways depending on whether the target population is known or not. Accordingly, two different procedures apply:

C.1.2 Procedure 1 – A biomechanical approach based on user group strength distributions

This procedure describes a way to determine force multipliers (Fo) for optional but well defined working populations in anonymous situations.

In this case force multipliers may be found by the following steps:

1) analyse a given work cycle and each of its technical actions;

2) get a set of 100 % of the maximum voluntary contraction (MVC) reference distribution functions for each technical action, i;

3) adjust all of the 100 % MVC_i reference distributions to the demographic profile (age and gender) of the envisaged user population (see EN 1005-3:2002, 4.2.1, Table 1);

4) determine percentile Force Limits $F_{L,i}$ (e.g. 15th percentile) for each technical action, i, allowing a majority (e.g. 85 %) to work at $F_{L,i}$-levels;

5) normalize actual Loads L_i by $F_{L,i}$ – this yields % MVC_i-values that are not exceeded by the majority selected (e.g. 85 %);

6) calculate an average $\%\,\overline{\mathrm{MVC}}$ -value integrating all technical actions of a work cycle (see also Table C.1):

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$$\%\overline{MVC} = \frac{1}{T}\sum \Delta t_i \cdot \%MVC_I$$

where

T	cycle time
Δt_i	duration of exposure to workload i;
$\%MVC_i$	%MVC-value under workload i.

7) find appropriate force multipliers Fo for each working cycle (see Figure C.2.)

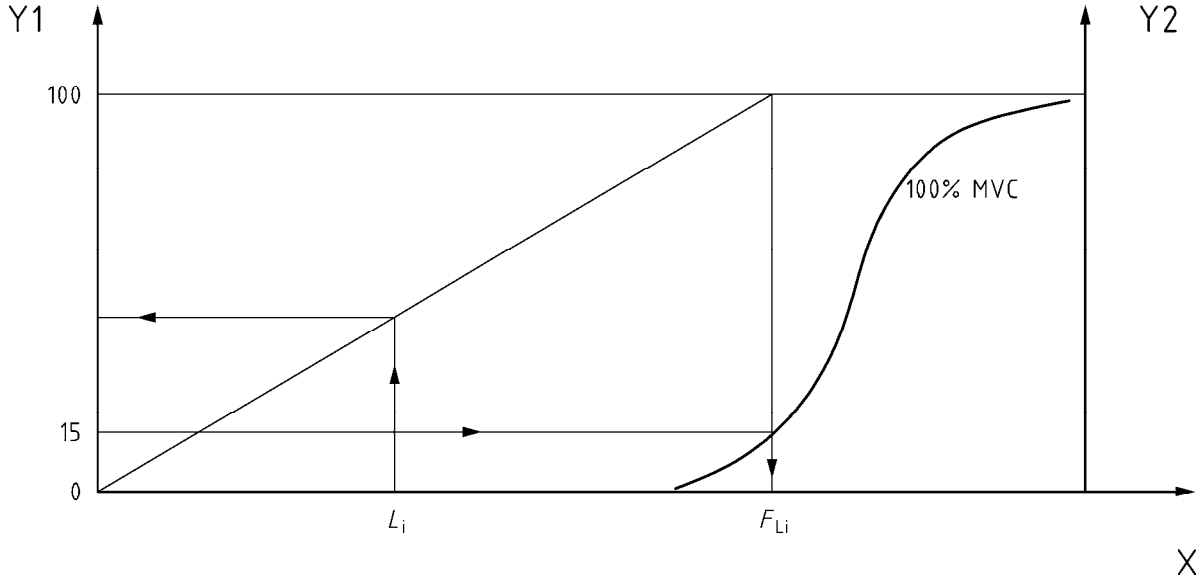

Figure C.1 — Finding % MVCi–values (step 1 to 5)

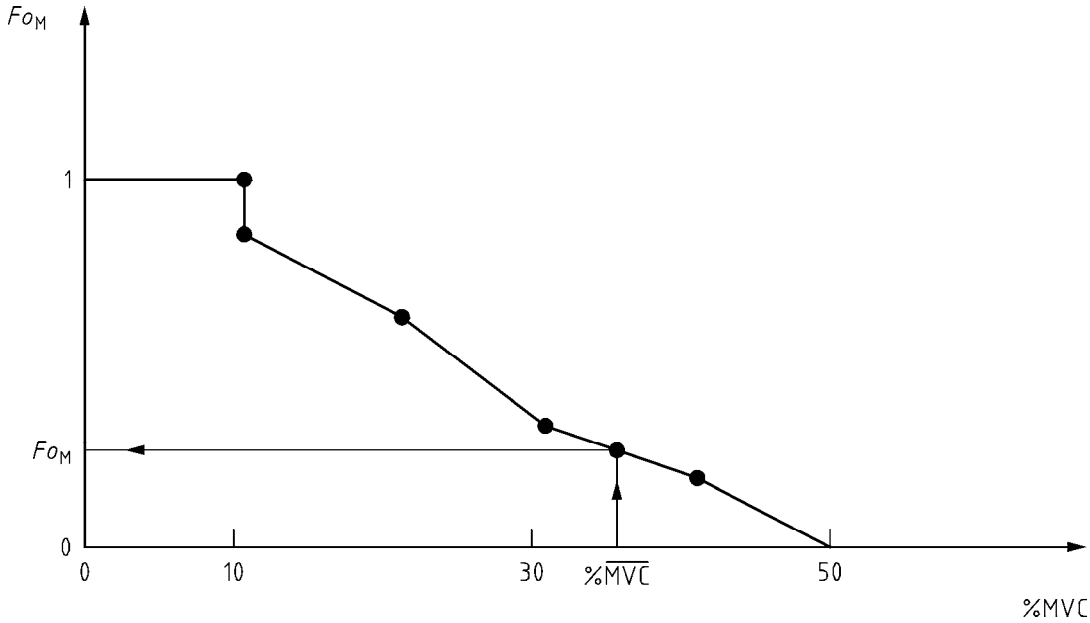

Figure C.2 — Finding force multiplier F_O (step 7)

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C.2 Procedure 2 – A psychophysical approach using the CR-10 Borg scale

It is requested to simulate the intended task by using a mock-up and a test group (see EN 614-1, 'evaluate task, evaluate with end users'). Applied forces may be estimated with end users by a specific scale proposed by Borg [8, 9] (Category Scale for the Rating of Perceived Exertion; CR-10 scale). This scale is a psychophysical tool that allows to describe and quantify the amount of muscular effort perceived by a subject performing a physical activity. The results of the implementation of CR-10 scale, when assessed by an adequate number of workers, have an accuracy, comparable to that of surface electromyography. The relationship between CR-10 scale result and exerted force (in percent maximum voluntary contraction - MVC) is: 10*CR -10 ≅ force (in percent) [15, 18, 36].

Quantification of the effort perceived by the whole upper limb should theoretically take place for every single technical action that makes up a cycle. For practical reasons, the technical actions that require minimal muscle involvement could be identified as 0,5 value in Borg's scale. Then the description procedure could only consider those actions, or groups of actions, that require more force than the minimal amount, always by using Borg's scale. Once this procedure has been carried out, the average weighted score for the whole of the cycle needs to be calculated.

Based on practical experiences, the following procedures are suggested:

The study on FORCE should come after that on technical action frequency: one needs already to know how the cycle works and especially the order and intensity of the successive force requirements inside a cycle.

The user needs to be asked whether inside the cycle there are technical actions that require muscle effort of the upper limbs. It is important to put the question in this way, because the worker often confuses muscle effort with the general tiredness that he/she feels at the end of the shift.

Once the actions, which imply the use of force, have been exemplified, the worker will be asked for rating between 0 and 10 on a scale form. The observer will ascribe the relevant duration to each of the strength exertions (in seconds and then as a % of the cycle duration). Since exposure assessment procedures are also intended to be preventive, it is important that the observer asks the worker to explain the reason for strength exertions. This is information of immediate practical interest because the presence of force when carrying out an action could be due to a technical defect in the product or tools used, or to a breakdown or a wrong choice of mechanical aids. Such problems are most often easily solvable.

Once the actions requiring force have been pinpointed and ranked according to Borg's scale, by ascribing to them the duration within the cycle, then all other technical actions in the remaining cycle time can be given the same score.

It is important that the worker him/herself does the scoring of the perceived physical effort in a given action. If this was done by an external observer, there would be major errors. In fact – and this is especially true of actions made by the smaller joints or for specific joint positions, such as pushing a button or a lever with the fingers, pinching etc. – the use of force is rarely perceivable by an external observer, even if (s)he is highly trained.

Once all information is obtained from the worker, any action requiring "PEAKS" (above 5 in Borg's scale) need to be recorded, and the average weighted score for all actions in the cycle needs to be calculated as reported in the example of Table C.1.

Table C.1 — Example of calculation of the average % MVC-value (procedure 1) and of the average score of perceived effort (procedure 2) considering all the technical actions in a work cycle lasting 35 s

Subdivision in time within a 35-second cycle	(A) % subdivision of the level of exertion in time	(B1) % of MVC or F_L	(B2) Borg scale score	AxB1 (%MVC or F_L)	AxB2 (perceived effort)
20 s	57 %	5	0,5	2,85	0,285
8 s	23 %	20	2	4,60	0,460
7 s	20 %	40	4	8,00	0,800
Final score				15,45	1,545

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Annex D
(informative)

Association between the OCRA index and the occurrence of Upper Limbs Work-related Musculoskeletal Disorders (UL-WMSDs): criteria for the classification of results and forecast models

D.1 General

The OCRA (OCcupational Repetitive Action) index is the ratio between the number of technical actions (performed during manual repetitive tasks) actually carried out during the work shift, and the number of reference technical actions which is specifically determined in the specific scenario [12, 33].

In practice is:

$$OCRA = a/b$$

where

a is the overall number of technical actions carried out in the shift;

b is the overall number or technical actions recommended in the shift.

The overall number of technical actions carried out within the shift is a known datum, which is calculated by organizational analysis (see also Annex A as a starting point).

The following general equation calculates the overall number of reference technical actions (RTA) within a shift:

$$RTA \frac{n}{n=1} \sum \left[CF \times \left(Fo_{Mj} \times Po_{Mj} \times Re_{Mj} \times Ad_{Mj}\right) \times D_j\right] \times Rc_M \times Du_M$$

where

n	is the number of repetitive tasks performed during shift;
j	is the generic repetitive task;
CF	is the “constant of frequency” of technical actions per minute = 30;
Fo_M; Po_M; Re_M; Ad_M	are multiplier factors with scores ranging from 0 to 1, in each of the (n) tasks;
D_j	is the net duration in minutes of each repetitive task;
Rc_M	is the multiplier factor for “lack of recovery period”;
Du_M	is the multiplier factor according to the overall duration of all repetitive tasks during a shift.

On the basis of recent studies [34] the association between the $OCRA$ index (independent variable) and the prevalence of Persons Affected (PA) by one or more UL- WMSDs (dependent variable) can be summarized by the following simple regression linear Equation (D.1):

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$$Y(PA) = 2{,}39 \pm 0{,}14\,(S.E.) \times OCRA \qquad \text{(D.1)}$$

where

$$Y(PA) = \frac{NPA \times 100}{NEP} \times 100$$

where

NPA is the Number of Person Affected by one or more UL-WMSDs;

NEP is the number of exposed individuals;

$S.E.$ is the Standard Error.

This regression equation is calculated without the constant (e.g., if $OCRA$ is 0, then no UL-WMSDs are supposed to be present), and starting from the study data examined until this moment, it has an adjusted R2 of 0,93 and it is extremely significant (p < 0,000 01) from a statistical point of view.

In this context the UL-WMSDs considered are all entrapment syndromes, tendonitis, peritendonitis of the upper limbs (shoulder included) confirmed by clinical examination and specific instrumental tests.

If the regression Equation (D.1) is being used as a forecast model, the OCRA index becomes a tool for forecasting the collective risk, for a given exposed population, to contract UL-WMSDs (in terms of PA) as shown in Table D.1.

Table D.1 — Forecast of PA (central tendency) for a group of exposed individuals, given specific OCRA index values

OCRA Value	Central
1	2,39 %
2	4,78 %
4	9,56 %
8	19,12 %

On another hand, also available data on the trends of PA in a reference working population never exposed to occupational risks for the upper limbs, are relevant for the purposes of this European Standard.

In a sample reference group of 749 subjects (310 males and 439 females) [11, 34], general and specific age and gender PA rates were computed. Considering the partial values of PA in different age and gender subgroups of this sample, it was possible to compute a standardized (for age and gender) rate (PA) with reference to the age and gender composition of a total national (Italian) workforce. Using statistical inferential procedures, the 90 % confidence limits and of the 5th and 95th percentiles of the standardized PA distribution were computed, as reported in the following Table D.2:

Table D.2 — PA values distribution as estimated in a working population never exposed to occupational risks for the upper limbs

Health effect	5 th percentile	50 th percentile central value	95 th percentile
PA	2,6	3,7	4,8

Using the PA variable in the reference not exposed population, $OCRA$ index reference limits were established starting from the 95 th percentile as the “driver value” for the so called green limit and from twice the 50th percentile as the “driver value” for the so called red limit.

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Those “driver” values of PA expected in the reference working population (not exposed) have been compared with the regression Equation (D.1) at the level corresponding to the 5th percentile (obtained using the $S.E.$): in such a way, by adopting a prudential criterion of assessment of not acceptable (yellow) or at risk (red) results, it was possible to found the $OCRA$ values corresponding respectively to the green and red limits and discriminating green, yellow and red areas as schematically shown in Figure D.1.

In practice:

— The green limit means that, at that level, in the exposed working population are forecasted, almost in 95 % of cases, PA values higher than the 95th percentile (PA = 4,8 %) expected in the reference (not exposed) population.

— The red limit means that, at that level, in the exposed working population are forecasted, almost in 95 % of cases, PA values higher than twice the 50th percentile (PA = 3,7 $\times$ 2 = 7,4 %) expected in the reference (not exposed) population.

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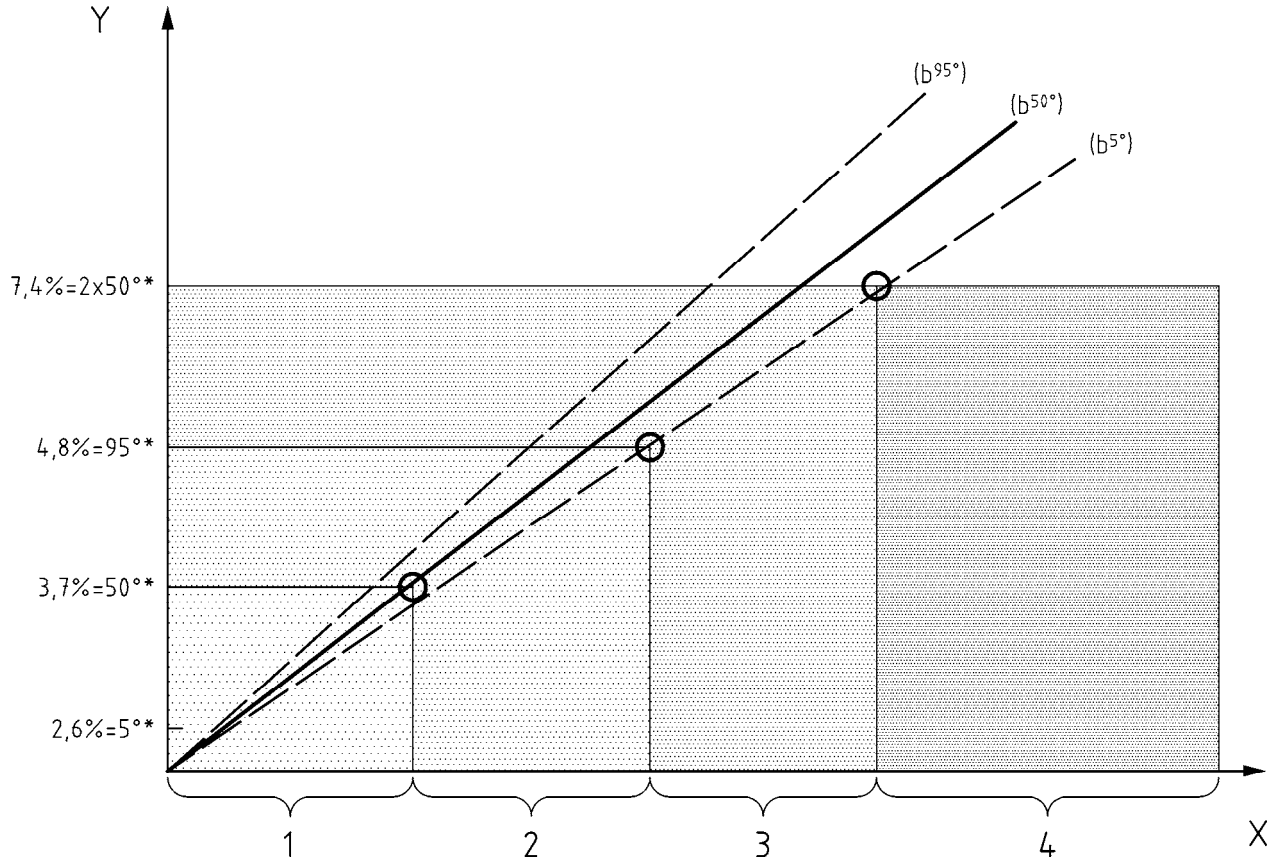

Key

X	OCRA	
Y	PA	
1	optimal, OCRA ≤ 1,5	
2	acceptable, OCRA ≤ 2,2	
3	borderline, OCRA ≤ 3,5	
4	Risk:	low ≤ 4,5, medium ≤ 9, high ≤ 9.

Figure D.1 — Schematic representation of the procedure adopted to define OCRA green and red limits based on PA in the reference population and using Equation D.1

D.2 OCRA Index values, exposure areas and consequent actions

Following the approach and using the data that have been synthetically presented, it becomes possible to identify the different risk areas (green, yellow and red) with “Key” *OCRA* values and to indicate the consequent preventive actions as reported in Table D.3.

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Table D.3 — Risk assessment criteria based on " key" OCRA index values

Zone	OCRA Value	Risk level	Consequences
Green	≤ 2,2	No risk UL-WMSDs (PA) forecast is not significantly different from the one expected in the reference population.	Acceptable No consequences
Yellow	2,3 to 3,5	Very low risk UL-WMSDs (PA) forecast is higher than previous but lower than twice the one expected in the reference population.	Advisable to set up improvements with regard to structural risk factors (posture, force, technical actions etc.); otherwise the existence of a "residual risk" that can be managed by "other organizational measures" should be mentioned in the user manual.
Red	> 3,5	Risk UL-WMSDs (PA) forecast is higher than twice the one expected in the reference population. The higher the index, the higher the risk.	Redesign of tasks and workplaces is recommended: if risk reduction to acceptable condition is not possible the existence of a "residual risk" that can be managed by "other organizational measures" should be mentioned in the user manual.

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Annex E
(informative)

Influence of recovery periods pattern and work time duration in determining the overall number of reference technical actions within a shift (RTA) and, consequently, the OCRA index

The procedure for counting the overall number of reference technical actions within a shift (RTA) suggested in Method 2 of this European Standard has been specifically established considering a common and usual organizational context in which a repetitive task duration lasts 240 min to 480 min per shift and two (plus the lunch break) breaks of 10 min (per shift) are provided: those two variables (task duration and breaks pattern) are generally out of the direct influence of the machinery designer.

In the general equation for computing the overall number of reference technical actions within a shift (RTA), "Recovery periods multiplier" Rc_M and "Duration of repetitive tasks multiplier" Du_M were consequently set at constant values of, respectively, 0,6 (Rc_M) and 1 (Du_M)

However, if the machinery designer can influence those two variables (for instance by giving specific information for use and recommendation for the intermediate and final user), the overall number of reference technical actions (RTA) can be increased by reducing repetitive task duration and/or providing more breaks and/or rotating on other tasks that can represent a recovery period for the upper limbs.

In this regard, in order to help the machinery designer, especially when applying Method 2, if a "yellow" condition results, the following criteria and computational tools are given:

a) Breaks and recovery period.
For repetitive task, the reference condition is represented by the presence, **for each hour of repetitive task**, of work breaks (during which one or more of the muscle groups are usually involved in the work task are basically inactive) of at least 10 min consecutively or in a ratio of 5:1 between work time and recovery periods [4, 12, 13, 33, 43].

In relation to these reference criteria it is possible to consider how many hours, during the shift, do not have an adequate recovery period. It requires the observation, one by one, of the single hours that make up a working shift: for each hour, a check needs to be made if there are repetitive tasks and if there are adequate recovery periods. For the hours preceding a meal break (if it is present), and for the hour before the end of the shift, the recovery period is considered to be represented by these two events. On the basis of the presence or absence of adequate recovery periods within every hour of repetitive work, the number of hours with "no recovery " is to be counted.

Once the number of hours without an adequate recovery period have been counted, it is possible to use the following Table E.1, for determining the proper values of the "Recovery periods multiplier" RcM to be used in the general equation for determining the overall number of reference technical actions (RTA).

Table E.1 — Recovery periods multiplier (RcM) in relation to the number of hours without an adequate recovery period

Number of hours without adequate recovery	0	1	2	3	4	5	6	7	8
Multiplier Rc_M	1	0,90	0,80	0,70	0,60	0,45	0,25	0,10	0

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b) Repetitive work overall daily duration.
Within a working shift, the overall duration of manual repetitive tasks is important to determine the overall risk for the upper limbs. When repetitive manual tasks last for a good part (240 min to 480 min) of the shift the "Duration of repetitive tasks multiplier" Du_M is equal to 1. In some contexts, however, there may be differences with respect to this more "typical" scenario (e.g., part-time work, repetitive manual task for only part of a shift); the multiplier (Du_M) considers these changes with respect to "usual" exposure conditions. Table E.2 gives the values of Du_M (to be used for recalculating the RTA), in relation with the overall daily duration of manual repetitive tasks [13, 30, 33].

Table E.2 — Elements for the determination of the Duration multiplier (DuM) in relation to the foreseen overall daily duration (in minutes) of manual repetitive tasks

Total time (in minutes) devoted to repetitive tasks during shift	< 120	120 to 239	240 to 480	> 480
Duration Multiplier Du_M	2	1,5	1	0,5

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Annex F
(informative)

An application example of risk reduction in a mono-task analysis

F.1 Foreword

When the result of the risk evaluation shows an unacceptable risk level, the designer should reduce it by optimising one or more of the following factors:

— number of technical actions needed in a work cycle;

— cycle time;

— awkward postures;

— level of force of the technical actions;

— additional factors.

He can also inform the user that it is possible to reduce the risk by reducing the task duration, adding breaks or introducing job rotation.

Before starting with an example, it could be useful to resume the time units proposed in the OCRA analysis:

— shift duration: minutes (min);

— cycle time: seconds (s) or hundredths of a minute (HM).The task designers in general use HM to describe the cycle time (seconds = HM $\times$ 0,6);

— technical action duration: the common task analysis methods, generally used in industry, propose HM as time units;

— technical action frequency: number of actions per minute.

F.2 General: technical characteristics of the task

The machinery designer should first determine:

— the foreseeable sequence and number of technical actions (for each upper limbs) needed to carry out a single cycle of the considered repetitive task (see Annex A);

— the foreseeable cycle time by eventually adopting one of the task analysis methods generally used in industry, as referred in Annex A..

This example describes the design of a task for the use of new machinery (in an assembly line) which consists of checking, at the end of the assembly-line, an electrical engine part via visual inspection, possible only by rotating the piece. The final operation is to store the pieces in a box. During a work cycle four pieces are checked.

To complete a cycle of four pieces using the machinery, the designer determines 21 technical actions for the right hand and 12 for the left, with a proposed cycle time of 20,5 s for the four pieces (or in 34,2 HM).

In general, during a work cycle one or more pieces can be handled.

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The action frequencies will be 61,36 actions per minute for the right and 35 actions per minute for the left upper limb. To calculate action frequency (for each upper limb) use the following equation:

$$\text{Technical action frequency} = \frac{\text{total number of actions to complete a cycle (for each arm)}}{\text{cycle time (s)}} \times 60$$

The technical actions necessary to complete a cycle, with the right hand, are: pull four pieces together, grasp one piece, turn it for a visual inspection, turn it again, repeat (grasp, turn, turn again), another three times for the other three pieces, take one piece, position it in the final container, take the second, position it, take the third, position it, take the last, position it (Table F.1).

Table F.1 — Identification of the technical actions in a cycle, for each upper limbs

Right upper limb	Number of technical actions-right	Left upper limb	Number of technical actions-left
pull four pieces together	1		
grasp piece (1st)	1	grasp piece (1st)	1
turn it for a visual inspection (1st)	1	turn it for a visual inspection (1st)	1
turn it again (1st)	1	turn it again(1st)	1
grasp piece (2nd)	1	grasp piece(2nd)	1
turn it for a visual inspection (2nd)	1	turn it for a visual inspection(2nd)	1
turn it again(2nd)	1	turn it again(2nd)	1
grasp piece (3srd)	1	grasp piece (3rd)	1
turn it for a visual inspection (3rd)	1	turn it for a visual inspection (3rd)	1
turn it again (3rd)	1	turn it again (3rd)	1
grasp piece (4th)	1	grasp piece (4th)	1
turn it for a visual inspection (4th)	1	turn it for a visual inspection (4th)	1
turn it again (4th)	1	turn it again (4th)	1
take (1st)	1		
position (1st)	1		
take (2nd)	1		
position (2nd)	1		
take (3rd)	1		
position (3rd)	1		
take (4th)	1		
position (4th)	1		
Total technical actions	**21**		**12**
Cycle time	**20,5 s (or 34,2 *HM*)**		**20,5 s (or 34,2 *HM*)**
Frequency actions/minute	**61,36**		**35**

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F.3 Hazard identification

The designer shall preliminarily perform a “hazard identification” to decide whether or not to proceed with a risk estimation. The “no hazard” option is present when:

— the task is not characterized by cycles;

— the task is characterized by cycles, but perceptual or cognitive activities are clearly present and upper limb movements are residual.

In the example, the machinery works with cycles with prevalent manual activity: the designer has to proceed with the risk assessment starting with.

F.4 Method 1

In the example, reported in F.1:

— the foreseeable frequency of the technical actions is 61 actions per minute.

— the presence of this condition implies the use of:

F.5 Method 2

F.5.1 Description of awkward postures and movements and evaluation of the corresponding Posture multiplier (Po_M)

With the proposed structure of the work place, the following awkward postures and movements have to be described for the different joints of both the upper limbs, as suggested in Table F.2.

To simplify the explanation, in this example we will only consider the right upper limb, but the designer has to analyse both.

Table F.2 — Awkward postures and movements estimation for the right upper limb

Right upper limb technical actions	Number of technical actions - right	Elbow awkward postures/ movements	Hand awkward postures/ movements
pull four pieces together	1		
grasp pieces (1st) (2nd) (3rd) (4th)	4		Palmar grip
turn them for a visual inspection (1st) (2nd) (3rd) (4th)	4	Flexion ≥ 60°	Palmar grip
turn them again (1st) (2nd) (3rd) (4th)	4	Flexion ≥ 60°	Palmar grip
Take (1st) (2nd) (3rd) (4th)	4	Flexion ≥ 60°	pinch
position (1st) (2nd) (3rd) (4th)	4	Flexion ≥ 60°	pinch
Total number of technical actions	21		

When the duration of each technical action and the distribution of the technical actions in the cycle is similar, it is possible to estimate the duration (in % of the cycle time) of an awkward posture or movement, dividing the number of technical actions found in that specific awkward posture or movements by the total number of technical actions, as described in Table F.3.

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When duration and distribution of each technical action in the cycle are different, it is more precise to estimate the duration (in % of the cycle time) dividing the duration (in hundredths of a minute = *HM*) of technical actions, found in a specific awkward posture or movement, by the total duration of the cycle time (in *HM*), as described in Table F.4.

A software (MIDA CEN) is available for calculating not only the % duration of the awkward posture or movements, but also the final *OCRA* index. By entering the durations of the actions (in *HM*) for each action and for awkward postures or movements (as proposed in Table F.4) the calculation is made.

Table F.3 — Estimation of the proportional duration, in a cycle time, of a joint in awkward posture or movements, dividing the number of technical actions, found in that specific awkward postures or movements by the total number of technical actions

Awkward postures/movements	Number of technical actions in awkward postures or movements	Total number of technical actions	Proportional duration
Elbow flexion-extension	16	21	76 %
Hand Palmar grip	12	21	57 %
Hand pinch	8	21	38 %

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Table F.4 — Evaluation of the proportional duration (% of cycle time) of a joint in awkward postures and movements, dividing the duration (*HM*) of technical actions found in that specific awkward posture or movements, by the total duration of the cycle time (*HM*)

Shoulder postures and movements			Elbow movements			Wrist postures and movements				Hand postures and movements					Additional factors					Technical actions of right upper limb		Description of upper limbs technical actions
Abduction > 45°	Flexion > di 70° to 80°	Extension > di 20°	Pronation > 60°	Supination > 60°	Flexion extension > 60°	Flexion > 45°	Extension > 45°	Radial deviation > 15°	Ulnar deviation > 20°	Power grip	Power grip with narrow span	Pinch	Palmar grip	Hook grip	Precision	Vibration	Compression	Strikes	Ripping movements	Duration (H.M.) of group of identical technical action	TOT – Technical action per cyle	right
																				1	1	Pull
													7,4							7,4	4	Grasp
					7,4								7,4							7,4	4	Turn
					7,4								7,4							7,4	4	Turn
					5,5							5,5								5,5	4	Take
					5,5							5,5								5,5	4	Place
					25,8							11	22,2							34,2	21	
					75							32	65							Proportional duration (%)		

NOTE Cycle time 34,2 *HM* equals 20,5 s.

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Using the results proposed in Table F.5, the Posture Multipliers (Po_M) are:

— for elbow in flexion/extension (≥ 60°) for 2/3 (from 51 % to 80 %) of the cycle time Po_M = 0,7;

— for hand in pinch and in palmar grip for 97 % of the cycle time Po_M = 0,5.

The Posture Multiplier Po_M, that represents the final posture evaluation, is the lowest score: Po_M = 0,5.

Table F.5 — Multiplier for awkward postures (Po_M)

Awkward posture [12]	Portion of the cycle time			
	Less than 1/3 from 1 % to 24 %	1/3 from 25 % to 50 %	2/3 from 51 % to 80 %	3/3 more than 80 %
Elbow supination (≥ 60°)	1	0,7	0,6	0,5
Wrist extension (≥ 45°) or flexion (≥ 45°)				
Hand pinch or hook grip or palmar grip (wide span)				
Elbow pronation (≥ 60°) or flexion/extension (≥ 60°)	1	1	0,7	0,6
Wrist radio-ulnar deviation (≥ 0°)				
Hand power grip with narrow span (≤ 2 cm)				

F.5.2 Repetitiveness multiplier (Re_M)

The task requires the performance of the same working movements for more than 50 % of the cycle time. In fact the sequence of the technical actions ”grasp, turn, turn” is repeated 4 time and lasts 22,2 HM, 65 % of the cycle time (Table F.4).

Repetitiveness multiplier Re_M will be: 0,7.

The software (MIDA CEN) enters the Repetitiveness multiplier (Re_M) in the OCRA index computation. This takes place by writing “yes” when it is present or “no” when repetitiveness does not occur.

F.5.3 Evaluation of average force level and the corresponding Force Multiplier (Fo_M)

The technical actions requiring force are showed in Table F.6. For each technical action (or group of identical actions) the following parameters are indicated:

— the duration (x);

— the proportion of its duration in the cycle (j = x/cycle time);

— the force level required, using the Borg scale (y) or the % of Fb or % of MVC (z).

By multiplying (y) x (j) or (z) x (j) and summing the results, the average force level will be obtained.

NOTE The result using the Borg scale is 0,9, using the % of MVC will be 9,49.

The data proposed in Table F.7 determine the Force multiplier (Fo_M) corresponding to the average force level estimated.

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The Fo_M equals 0,88 (interpolated value).

The software (MIDA CEN) calculates the average force level and inserts the corresponding Force Multiplier (Fo_M) in the $OCRA$ index computation by inserting for each technical actions (or group with identical actions) the duration (in HM) and the corresponding score (% of **Fb** or Borg-CR-10 score).

Table F.6 — Evaluation of average force level (Force Level)

Description of upper limbs technical actions	Technical actions of right upper limb		Force		% tot. cycle time	Sorce by Borg	Sorce by % F_b
right	Technical actions per cycle	duration (in HM)	Borg scale score	% F_b			
equation		x	y	z	$j = y/35$	j x j	j x z
Pull	1	1	2	20	0,03	0,06	0,64
Grasp	4	7,4	0,5	5	0,22	0,11	1,08
Turn	4	7,4	0,5	5	0,22	0,11	1,08
Turn	4	7,4	0,5	5	0,22	0,11	1,08
Take	4	5,5	0,5	5	0,16	0,08	0,80
Place	4	5,5	3	30	0,16	0,48	4,81
Total	21	34				0,95	9,49

Table F.7 — Multiplier relative to the different use of force (Fo_M)

Force level in % of MVC or MC	5	10	20	30	40	≥ 50
CR-10 Borg Score	0,5 very, very weak	1 very weak	2 weak	3 moderate	4 somewhat strong	≥ 5 strong/very strong
Force Multiplier (Fo_M)	1	0,85	0,65	0,35	0,2	0,01

F.5.4 Determination of the Recovery period multiplier (Rc_M) and the Duration multiplier (Du_M)

The repetitive task duration and the breaks distribution is not under the control of the machinery designer.

Since these elements are included in the $OCRA$ index, the designer can use the two corresponding multipliers referring them to a standard shift duration of 480 min, with a meal break and two breaks of 10 min each, one before and the other after the meal break. The net duration of repetitive task (D) in this case will be 460 min.

Considering the data exposed in Tables F.8 and F.9 the reference multipliers will be (see also Annex E).

— Recovery period multiplier: Rc_M = 0,60 (corresponding to 4 h without an adequate recovery period).

— Duration multiplier: DuM = 1 (corresponding to a net repetitive task duration of 240 min to 480 min).

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Table F.8 — Elements for the determination of the Recovery period multiplier (Rc_M)

Number of hours without adequate recovery	0	1	2	3	4	5	6	7	8
Recovery Multiplier Rc_M	1	0,90	0,80	0,70	0,60	0,45	0,25	0,10	0

Table F.9 — Elements for the determination of the Duration Multiplier (Du_M)

Total time (in minutes) devoted to repetitive tasks during shift	< 120	120 to 239	240 to 480	> 480
Duration Multiplier Du_M	2	1,5	1	0,5

F.5.5 Computation of reference technical actions per minute (RF)

$$RF = CF \times Po_M \times Re_M \times Ad_M \times Fo_M \times (Rc_M \times Du_M)$$

where

CF is the “constant of frequency” of technical actions per minute = 30;

Po_M; Re_M; Ad_M; Fo_M are multipliers for the risk factors postures, repetitiveness, additional, force;

Rc_M is the multiplier for the risk factor “lack of recovery period”;

Du_M is multiplier according to the overall duration of repetitive task(s) during a shift.

In the presented example:

$$RF = 30 \times 0{,}5 \times 0{,}7 \times 1 \times 0{,}88 \times (0{,}6 \times 1) = 5{,}54$$

F.5.6 Computation of the OCRA index

$$OCRAindex = \frac{FF}{RF}$$

where

FF is the Foreseeable Frequency per minute of technical actions needed to carry out the task;

RF is the Reference Frequency per minute of technical actions.

In the presented example:

$$OCRA\ index = 61{,}36 / 5{,}54 = 11{,}1$$

F.5.7 OCRA index calculation for mono task analysis when the repetitive task duration should be assessed

When designing a machinery the task duration should be assessed. The OCRA index is given by the following general equation:

$$OCRA = \frac{ATA}{RTA}$$

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where

ATA is the overall number of Actual Technical Actions needed in the shift;

RTA is the overall number of reference technical actions in the shift.

a) Computation of the overall number technical actions within a shift (ATA)

The overall number of Actual Technical Actions (ATA) actually carried out within the shift for a repetitive task can be calculated by multiplying FF (frequency of technical actions per minute) with D (net duration of the repetitive task in minutes).

$$ATA = FF \times D$$

In the example:

$$ATA = 61{,}36 \times 460 = 28224$$

b) Computation of the overall number of reference technical actions within a shift (*RTA*)

The following general equation calculates the overall number of reference technical actions within a shift (RTA):

$$RTA = CF \times Po_M \times Re_M \times Ad_M \times Fo_M \times (D \times Rc_M \times Du_M)$$

where

CF is the “constant of frequency” of technical actions per minute (CF = 30 per minute);

Po_M; Re_M; Ad_M; Fo_M are multipliers for the risk factors postures, repetitiveness, additional, force;

D is the net duration of repetitive task, in minutes;

Rc_M is the multiplier for the risk factor “lack of recovery period”;

Du_M is the multiplier according to the overall duration of repetitive task(s) during a shift.

In the present example where D = 460 min:

$$RTA = 30 \times 0{,}5 \times 0{,}7 \times 1 \times 0{,}88 \times (460 \times 0{,}6 \times 1) = 2550$$

c) Risk index computation: $OCRA$ Index for mono task analysis considering the repetitive task duration

The $OCRA$ Risk Index is obtained by comparing, for each upper limb, the overall number of technical actions carried out in the shift (ATA) with the overall number of reference technical actions within a shift (RTA), by means of this equation:

$$OCRA = \frac{ATA}{RTA}$$

In the example the risk evaluation leads to a risk condition (red zone).

$$OCRA\ Index = \frac{28224}{2550} = 11{,}1 \text{ (Table F.10)}$$

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Table F.10 — A simple model to calculate the mono task *OCRA* index

General information	
Shift Duration	480
Breaks (min)	20
Non repetitive work time (min)	0
Work time considered as recovery (min)	0
Repetitive work net time	460
Number of cycles per shift	1 344
Number of hours without an adequate recovery period (h)	4
Recovery Multiplier (RC_M)	0,6
Constant of Frequency (CF)	30
right	
Force Multiplier (FO_M)	0,88
Posture Multiplier (PO_M)	0,5
Additional factors Multiplier (Ad_M)	1
Repetitiveness multiplier (Re_M)	0,7
(Reference number of technical actions without RC_M)	4 250,4
Reference number of technical actions (RTA)	2 550,24
Cycle time	**20,5**
Total number of Actual Technical Actions observed (ATA)	28 224
Frequency (number of technical actions/min)	61,4
Number of technical actions in the cycle	21
Duration Multiplier (DU_M)	1
***OCRA* Index**	**11,1**

F.5.8 Solutions to reduce the risk level

Now the machinery designer knows that the use of machinery, proposing a cycle time of 21 s, produces an *OCRA* index equal to 11,1 (red zone = risk present).

He can use different solutions to reduce the risk:

a) Increase cycle time duration

Reducing the number of cycles and increasing consequently the cycle time, as shown in Table F.11 and Table F.12, the designer is proposing to significantly reduce the production.

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Table F.11 — *OCRA* index: example of reduction of the number of pieces, increasing the cycle time, to obtain a yellow and a green area

Number of cycles for shift	Frequency	Cycle time *s*	OCRA index
1 344	61,2	20,5	11,1 (Red)
430	19,6	64	3,5 (Yellow)
270	12,3	102,2	2,2 (Green)

Table F.12 — *OCRA* index: example of reduction of the number of pieces to obtain results in yellow and green zones, increasing the cycle time

	Example	Reduction
Shift duration	480	480
Breaks (min)	20	20
Non repetitive work time (min)	0	0
Work time considered as recovery (min)	0	0
	460	460
Write X for each task analysed	x	x
Repetitive work net time	460	460
Number of cycles per shift	**430**	**270**
Number of hours without an adequate	4	4
Recovery period (h)	0	0
Recovery Multiplier (Rc_M)	0,6	0,6
Constant of Frequency (CF)	30	30
right		
Force Multiplier (Fo_M)	0,88	0,88
Posture Multiplier (Po_M)	0,5	0,5
Additional factors Multiplier (Ad_M)	1	1
Repetitiveness Multiplier (Re_M)	0,7	0,7
Reference number of technical actions without Rc_M	4 250,4	4 250,4
Reference number of technical actions (RTA)	2 550,24	2 550,24
Cycle time	**64,2**	**102,2**
Total number of Actual Technical Actions observed (ATA)	9 030	5 670
Frequency (number of technical actions/min)	19,6	12,3
Number of technical actions in the cycle	21	21
Duration Multiplier (Du_M)	1	1
***OCRA* index**	**3,5**	**2,2**

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b) Reduction of the technical actions

To avoid a task creating a risk, the designer needs to reduce the technical actions numbers optimising their distribution and introducing more automation, without reducing the production.

As an example, the four pieces can arrive automatically at the workstation and a new mechanical device can allow the worker to control two pieces at the same time. A pneumatic aid can be introduced to lift together the four pieces at the end of the task, for putting them in the final container: it reduces both the number of actions and the use of force.

The technical actions necessary to complete a cycle, with the right hand, will be now: grasp two pieces together, turn them for a visual inspection, turn them again, grasp the last two pieces, turn them for a visual inspection, turn them again, take the mechanical aid, position the aids on four pieces, position the aid on the container, release the four pieces. The number of actions will be now 9 (Table F.13) for each upper limb.

Table F.13 — Identification of the technical action in a cycle

Right upper limb	**Number of technical actions – right**	**Left upper limb**	**Number of technical actions – left**
grasp 2 pieces (1st) (2nd)	1	grasp piece (1st)	1
turn item for a visual inspection (1st) (2nd)	1	turn it for a visual inspection (1st)	1
turn item again (1st) (2nd)	1	turn it again (1st)	1
grasp the last 2 pieces (3rd) (4th)	1	grasp piece (2nd)	1
turn item for a visual inspection (3rd) (4th)	1	turn it for a visual inspection (2nd)	1
turn item again (3rd) (4th)	1	turn it again (2nd)	1
position the aids on 4 pieces	1		1
position the aid on the container	1		1
release the 4 pieces	1		1
Total of technical actions	**9**		**9**
Cycle time	**20,5 s (or 34,2 *HM*)**		**21**
Frequency/minute	**25,7**		**25,7**

By introducing an aid, the force is now practically absent (Table F.14).

The average level of force, using the Borg scale is 0,22, using the % of Fb will be 3,13.

The data proposed in Table F.7 enables the designer to find the Force Multiplier corresponding to the average force level estimated: the Fo_M is 1 (absence of force).

The scheme, proposed in Table F.15 (MIDA CEN) allows the designer to recalculate not only the % duration of the awkward posture or movements, but also the final $OCRA$ index. This is achieved by entering the new actions duration (in HM) for each action and then for those in awkward postures or movements.

According to the Table F.5, the new Posture Multipliers (Po_M) are:

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— elbow in flexion/extension for 1/3 of the cycle time Po_M = 1

— palmar grip 1/3 of the cycle time Po_M = 0,7

The Posture Multiplier Po_M that represents the final posture evaluation is the lowest score Po_M = 0,7

Table F.14 — The evaluation of the new (b) average force level (Force multiplier)

Description of upper limbs technical actions	Technical actions of right upper limb		Force		% tot. cycle time	Score by Borg	Score by % F_b
right	Technical actions per cycle	duration (in HM)	Borg scale score	% F_b			
Grasp	2	5,0	0,5	5	0,15	0,07	0,73
Turn	2	5,0	0,5	5	0,15	0,07	0,73
Turn	2	5,0	0,5	5	0,15	0,07	0,73
Position aid	1	6,4	0,0	5	0,19	0,00	0,94
Position aid	1	6,4	0,0	0	0,19	0,00	0,00
Release aid	1	6,4	0,0	0	0,19	0,00	0,00
Total	9	28				0,22	3,13

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Table F.15 — Estimation of the new (*b*) proportional duration, in a cycle time, of a joint in awkward postures or movements, dividing the duration (in *HM*) of technical actions found in that specific awkward posture or movement, by the total duration of the cycle time (in HM)

Shoulder postures and movements			Elbow movements			Wrist postures and movements				Hand postures and movements					Additional factors					Technical actions of right upper limb		Description of upper limbs technical actions
Abduction > 45°	Flexion > di 70° to 80°	Extension > di 20°	Pronation > 60°	Supination > 60°	Flexion extension > 60°	Flexion > 45°	Extension > 45°	Radial deviation > 15°	Ulnar deviation > 20°	Power grip	Power grip with narrow span	Pinch	Palmar grip	Hook grip	Precision	Vibration	Compression	Strikes	Ripping movements	Duration (H.M.) of group of identical technical action	TOT – Technical action per cycle	right
													5							5,0	2	Grasp
					5								5							5,0	2	Turn
					5								5							5,0	2	Turn
																				6,4	1	Position aid
																				6,4	1	Position aid
																				6,4	1	Release aid
					10								15							34,2	9	
					29								44									%
NOTE Cycle time 32,2 HM equals 20,5 s.																						

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The task now does not require the performance of the same working gestures for more than 50 % of the cycle time. In fact, the technical actions “grasp, turn, turn” are repeated only twice and last 15 in HM (44 % of the cycle time).

Repetitiveness multiplier will be: Re_M = 1.

In the presented example, the cycle time is again 20,5 s, maintaining the same production, but the frequency of the technical actions is now only 26,3 actions per minute (Table F.16):

$$ATA \times 26{,}3 \times 460 = 12098$$

$$RTA = 30 \times 1 \times 0{,}7 \times 1 \times 1 \times (460 \times 0{,}6 \times 1) = 5796$$

$$OCRA\ index = \frac{12098}{5796} = 2{,}1$$

Table F.16 — *OCRA* index in the redesigned workplace: the result is in green zone maintaining the same production

General information	
Shift duration	480
Breaks (min)	20
Non repetitive work time (min)	0
Work time considered as recovery (min)	0
Repetitive work net time	460
Number of cycles per shift	**1 344**
Number of hours without an adequate recovery period (h)	4
Recovery Multiplier (RC_M)	0,6
Constant of Frequency (CF)	30
right	
Force Multiplier (FO_M)	1
Posture Multiplier (PO_M)	0,7
Additional Multiplier (Ad_M)	1
Repetitiveness Multiplier (Re_M)	1
(Reference number of technical actions without RC_M)	9 660
Reference number of technical actions (RTA)	5 796
Cycle time	**20,5**
Total number of Actual Technical Actions observed (ATA)	12 096
Frequency (number of technical actions/min)	26,3
Number of technical actions in the cycle	9
Duration Multiplier (DU_M)	1
***OCRA* Index**	**2,1**

The designer, using the software, can also observe what happens by increasing production.

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By increasing the number of cycles in the shift by 1 344 to 1 700, the *OCRA* risk index will be 2,6, (yellow area). However, by changing the job organization (for example adding 2 other breaks of 10 min each) the *OCRA* risk index will be 2,1 (green zone) (Table F.17).

Table F.17 — *OCRA* index in the redesigned workplace: the results in green an in yellow area increasing the production and adding breaks

	Example	Reduction
Shift duration	480	480
Breaks (min)	40	40
Non repetitive work time (min)	0	0
Work time considered as recovery (min)	0	0
	460	440
Write X for each task analysed	x	x
Repetitive work net time	460	440
Number of cycles per shift	**1700**	**1700**
Number of hours without an adequate	4	2
Recovery period (h)	0	0
Recovery Multiplier (Rc_M)	0,6	0,8
Constant of Frequency (CF)	30	30
right		
Force Multiplier (Fo_M)	1	1
Posture Multiplier (Po_M)	0,7	0,7
Additional Multiplier (Ad_M)	1	1
Repetitiveness Multiplier (Re_M)	1	1
Reference number of technical actions without Rc_M	9 660	9 240
Reference number of technical actions (RTA)	5 796	7 392
Cycle time	**16,2**	**15,5**
Total number of Actual Technical Actions observed (ATA)	15 300	15 300
Frequency (number of technical actions/min)	33,3	34,8
Number of technical actions in the cycle	9	9
Duration Multiplier (Du_M)	1	1
***OCRA* index**	**2,6**	**2,1**

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c) Conclusion

The designer has to review the development of a machine that gives rise to hazards and high *OCRA*-values.

Primarily he has to consider if:

— the observed awkward postures and movements are strictly needed. It needs to be the first duty of the designer, at this stage of the development of a machine, to try to eliminate those postures or replace them by harmless ones considering the contents of the other CEN standards such as EN ISO 14738 or EN 1005-4;

— the parts of the equipment requiring the use of unsuitable handgrips can be redesigned as well as the forces to be used (EN 1005-3);

— the number of technical actions and the pace of the machinery can be re-engineered.

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Annex G
(informative)

Definition and quantification of additional risk factors

Apart the main risk factors (frequency and repetitiveness of technical actions, use of force, awkward postures and movements, lack of recovery periods, daily repetitive task duration) which are examined elsewhere, there are other factors, always of an occupational nature, that, according to suggestions by the literature, need to be taken into consideration when exposure is assessed [11, 12, 13]. They are defined here as additional. This is not because they are of secondary importance, but because each one of them can, from time to time, be present or absent in the contexts examined.

The list of these factors is not necessarily exhaustive and includes:

a) the use of vibrating tools (even if only for part of the actions);

b) requirement for precision of placement (tolerance 1 mm to2 mm in positioning a piece or object);

c) localized compressions on anatomical structures of the hand or of the forearm with tools, objects, or working areas;

d) exposure to cold or refrigeration;

e) the use of gloves which interfere with the handling ability required by the task;

f) objects handled have a slippery surface;

g) sudden movements, or “tearing”, “ripping” movements, or fast movements are required;

h) the technical actions required imply a counter shock (such as e.g. hammering, or hitting with a pick over hard surfaces, using the hand as a tool etc.).

NOTE This list is only concerned with factors of a physical or mechanical nature.

Other factors, which are listed under the general term of “psycho-social”, have also been called into play for determining the onset of Upper Limbs Work-related Musculo-Skeletal Disorders (UL-WMSDs). Among them, some are concerned with the individual, and cannot therefore be included in general methods considering a collective and occupational type of exposure of a target group.

The description of additional factors can take place in parallel with that of technical actions or of postures and movements.

For each of the physical-mechanical factors, it is necessary to specify for how much time (as a portion of the cycle/task time like 1/3, 2/3, 3/3) the factor is present, or to describe the frequency of occurrence of actions where that factor is present (especially for sudden movements and movements with counter shocks). A partial exception is represented by the factor defined as “vibrations” transmitted to the hand-arm system, that here is considered only to be present or not (for a fraction of the cycle and task time), but for which a detailed exposure assessment is due by other EN ISO standards [2, 3], or by European directive 2002/44/EG.

The assessment of additional factors begins with a definition of optimum conditions, as represented by the absence, or by the very limited presence, of additional factors: in this scenario the Additional factors Multiplier Ad_M equals 1; any discrepancy with respect to this optimal condition represents a contribution of additional factors to the overall exposure level, which grows with the growing portion of the cycle time during which additional factors (one or more) are present.

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In those cases the Additional factor Multiplier Ad_M equals:

— 0,95 if one or more additional factors are present at the same time for 1/3 (from 25 % to 60 %) of the cycle time;

— 0,90 if one or more additional factors are present at the same time for 2/3 (from 61 % to 80 %) of the cycle time;

— 0,80 if one or more additional factors are present at the same time for 3/3 (more than 80 %) of the cycle time.

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Annex H
(informative)

Risk assessment by Method 2 when designing " multitask" jobs

H.1 OCRA index calculation when two or more repetitive tasks should be assessed

When, designing a machinery, two or more repetitive tasks should be assessed together, for instance considering (or addressing information for the users to) the possibility of rotation on different tasks in different workstations of the machinery or along an assembly line, the risk assessment of repetitive handling at high frequency could be performed by Method 2, using the following general equation for calculating OCRA index:

$$OCRAindex = \frac{ATA}{RTA}$$

In practice, for jobs with two or more repetitive tasks, the $OCRA$ index is given by the ratio, for each upper limb, between the overall number of technical actions needed in the shift (ATA) and the overall number of reference technical actions in the shift (RTA).

The overall number of Actual Technical Actions needed (ATA) to perform the different repetitive tasks, is given by the following equation:

$$ATA = \sum_{J=1}^{n} \left(FF_j \times D_j \right)$$

where

n is the number of repetitive task/s performed during the shift;

D_j is the foreseeable net Duration (in minutes) of the task j;

FF_j is the Foreseeable Frequency of actions per minute of task j.

The following general equation calculates the overall number of reference technical actions within a shift (RTA):

$$RTA = \sum_{J=\cdots 1}^{n} \left[CF \times \left(Fo_{Mj} \times Po_{Mj} \times Re_{Mj} \times Ad_{Mj} \right) \times D_j \right] \times \left(Rc_M \times Du_M \right)$$

where

n is the number of repetitive tasks performed during the shift;

j is the generic repetitive task;

CF is the "constant of frequency" of technical actions per minute = 30;

Fo_{Mj}; Po_{Mj}; Re_{Mj}; Ad_{Mj} are multipliers for the risk factors postures, repetitiveness, additional, force, in each j repetitive task;

Dj is the foreseeable net duration (in minutes) of the repetitive task j;

Rc_M is the multiplier for the risk factor "lack of recovery period";

Du_M is the multiplier according to the overall duration of all repetitive tasks during a shift.

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In practice, to determine the overall number of reference technical actions within a shift (RTA), proceed as follows, by using the same procedures and data given in points from 5.3.4.1.2 to 5.3.4.1.6:

a) for each repetitive task, start from CF (30 actions/min);

b) for each repetitive task, CF (the frequency constant) has to be weighted (by the respective multipliers) considering the presence and degree of the following risk factors: force (Fo_M), posture (Po_M), repetitiveness (Re_M) and additional (Ad_M);

c) the weighted frequency so obtained, has to be multiplied, for each task, by the number of minutes of the foreseeable duration (D) of each repetitive task;

d) sum up the values obtained for the different tasks;

e) the resulting value has to be multiplied by the multiplier for lack of recovery periods (Rc_M) (see 5.3.4.1.6 and Annex E);

f) apply the last multiplier that considers the total time (Σ Dj) spent in repetitive tasks during the whole shift (Du_M) (see 5.3.4.1.6 and Annex E).

The value thus obtained represents the total number of reference technical actions (RTA) in the shift for the examined job (made up of two or more repetitive tasks).

Table 3 (in the standard) supplies the relevant values of the $OCRA$ index to assess the risk in relation to the 3-zone rating system (green, yellow, red) and to decide for consequent actions to be taken.

H.2 An application example: assessing repetitive tasks at a machine

H.2.1 Description of characteristics of two tasks

Two different workstations at a machine have to be used to produce mechanical components, each requiring a different number of technical actions and different duration of the cycle time. The main characteristics of the two different tasks A (see also Table H.3) and B (see also Table H.4) are synthesized below:

Table H.1 — Description of characteristics of two tasks

Main characteristic	Task A	Task B
Number of technical actions (right upper limb)	21	32
Number of technical actions (left upper limb)	8	16
Cycle time duration in seconds (and in HM)	30 (50)	60 (100)
Frequency of technical actions (right upper limb)	42	32
Frequency of technical actions (left upper limb)	16	16
Force (right upper limb)	very low	very low
Force (left upper limb)	absent	absent
Awkward postures and movements (right upper limb)	76 % in pinch and palmar grip 44 % elbow flexion/extension	24 % pinch
Awkward postures and movements (left upper limb)	40 % pinch 40 % elbow flexion/extension	24 % pinch
Additional factors	absent	absent
Repetitiveness (right upper limb)	present	absent
Repetitiveness (left upper limb)	absent	absent

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H.2.2 Definition of the corresponding multipliers

Considering the above characteristics, the corresponding multipliers for the right and for the left upper limb are (see also Tables H.3 and H.4):

Table H.2 — Definition of the corresponding multipliers

MULTIPLIERS	Task A		Task B	
	right	left	right	left
Force Multiplier (Fo_M)	1	1	1	1
Posture Multiplier (Po_M)	0,6	0,7	0,7	0,7
Additional Multiplier (Ad_M)	1	1	1	1
Repetitiveness Multiplier (Re_M)	0,7	1	1	1

H.2.3 Mono- task analysis separately for task A and B: computation of the overall number Actual Technical Actions (ATA) in task A (Table H.3) and task B (Table H.4)

Consider a scenario in which the two tasks are performed by two workers and that each of them executes only one task for the overall shift time: 460 min (D) and there are 2 breaks of 10 min and a lunch break.

In task A, choosing a cycle time lasting 30 s, the production will be 920 units in 460 min; in task B, with a cycle time of 60 s, the production will be 920 units in 460 m.

The overall number of Actual Technical Actions (ATA) carried out within the shift for task A and B can be calculated, multiplying FF (frequency of technical actions per minute) by D (net duration of the repetitive task in minutes).

$$ATA = FF \times D$$

In the example

ATA (task A: right upper limb) = 42 $\times$ 460 = 19 320

ATA (task B: right upper limb) = 32 $\times$ 460 = 14 720

ATA (task A: left upper limb) = 16 $\times$ 460 = 7 360

ATA (task B: left upper limb) = 16 $\times$ 460 = 7 360

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Table H.3 — Characteristics of task A: technical actions and their duration, postures and movements, force, additional factors and repetitiveness (derived by MIDA-CEN software for multitask analysis)

Shoulder postures and movements				Elbow movements				Wrist postures and movements					Hand postures and movements					Repetitiveness multiplier RE_M yes =1, No = 0	Additional factors								Force		Technical actions of right upper limb		Description of upper limbs technical actions
Abduction > 45°	Flexion > di 70° - 80°	Extension > di 20°	Other postures of movements	Pronation > 60°	Supination > 60°	Flexion extension > 60°	Other postures or movements	Flexion > 45°	Extension > 45°	Radial deviation > 15°	Ulnar deviation > 20°	Fine movements	Grip with narrow span	Pinch	Palmar grip	Hook grip	Fine movements		Precision	Vibration	Compression	Strikes	Ripping movements	Other additional factors	Other additional factors	Other additional factors	Borg scale scores	Force duration (HM)	Technical actions duration (cm/min)	TOT technical actions per cycle	**right**
						2												1									2	2	2	1	Pull
															8												0,5	8	8	4	Grasp
															10												0,5	10	10	4	Turn
																											0,5	10	10	4	Push
						10								10													0	10	10	4	Take
						10								10													0	10	10	4	Place
22													20		18			1										0,36	50	21	
				Posture Factor 0,60									Repetitiveness 0,7						Additional factors 1,00								Force factor 1,00				
																															left
						10								10													0,5	10	10	4	Take
						10								10													0,5	10	10	4	Place
20													20															0,20	20	8	
				Posture Factor 0,70									Repetitiveness 1						Additional factors 1,00								Force factor 1,00				

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Table H.4 — Characteristics of task B: technical actions and their duration, postures and movements, force, additional factors and repetitiveness (derived by MIDA-CEN software for multitask analysis)

Shoulder postures and movements				Elbow movements				Wrist postures and movements					Hand postures and movements					Repetitiveness multiplier RE_M yes =1, No = 0	Additional factors								Force		Technical actions of right upper limb		Description of upper limbs technical actions
Abduction > 45°	Flexion > di 70° - 80°	Extension > di 20°	Other postures of movements	Pronation > 60°	Supination > 60°	Flexion extension > 60°	Other postures or movements	Flexion > 45°	Extension > 45°	Radial deviation > 15°	Ulnar deviation > 20°	Fine movements	Grip with narrow span	Pinch	Palmar grip	Hook grip	Fine movements		Precision	Vibration	Compression	Strikes	Ripping movements	Other additional factors	Other additional factors	Other additional factors	Borg scale scores	Force duration (HM)	Technical actions duration (cm/min)	TOT technical actions per cycle	**right**
														12															12	8	Take
														12															12	8	Position
																													12	8	Take
																													12	8	Position
24																													48	32	
				Posture Factor 0,70									Repetitiveness 1						Additional factors 1,00								Force factor 1,00				
																															left
														12															12	8	Take
														12															12	8	Position
24																													24	16	
				Posture Factor 0,70									Repetitiveness 1						Additional factors 1,00								Force factor 1,00				

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H.2.4 Mono- task analysis: computation of the overall number of reference technical actions within a shift (RTA) in task A (Table H.5) and task B (Table H.6)

The following general equation calculates the overall number of reference technical actions within a shift (RTA) in task A and task B:

$$RTA = CF \times Po_M \times Re_M \times Ad_M \times Fo_M \times (D \times Rc_M \times Du_M)$$

where

CF	is the “constant of frequency” of technical actions per minute = 30;
Po_M; Re_M; Ad_M; Fo_M	are multipliers for the risk factors postures, repetitiveness, additional, force;
D	is the net duration in minutes of repetitive task=460;
Rc_M	is the multiplier for the risk factor “lack of recovery period”= 0,6 (see Annex E);
Du_M	is the multiplier according to the overall duration of repetitive task (s) during a shift = 1 (see Annex E).

RTA (task A: right upper limb) = 30 × 0,6 × 0,7 × 1 × 1 × (460 × 0,6 × 1) = 3477,6

RTA (task B: right upper limb) = 30 × 0,7 × 1 × 1 × 1 × (460 × 0,6 × 1) = 5796

RTA (task A: left upper limb) = 30 × 0,7 × 1 × 1 × 1 × (460 × 0,6 × 1) = 5796

RTA (task B: left upper limb) = 30 × 0,7 × 1 × 1 × 1 × (460 × 0,6 × 1) = 5796

H.2.5 Mono- task analysis: computation of the OCRA index in task A (Table H.5) and task B (Table H.6)

In the present example:

$OCRA$ index (task A: right upper limb) = 19 320/3 477,6 = 5,6 (red zone)

$OCRA$ index (task B: right upper limb = 14 720/5 796 = 2,5 (yellow zone)

$OCRA$ index (task A: left upper limb) = 7 360/5 796 = 1,3 (green zone)

$OCRA$ index (task B: left upper limb) = 7 360/5 796 = 1,3 (green zone)

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Table H.5 — *OCRA* index in task A (net duration of the repetitive task equal to 460 min)

	right	left
Shift duration	480	
Breaks (min)	20	
Non repetitive work time (min)	0	
Work time considered as recovery (min)	0	
	460	
Write X for each task analysed	x	x
Repetitive work net time	460	
Number of cycles per shift	**920**	
Number of hours without an adequate	4	
Recovery period (h)	0	
Recovery Multiplier (Rc_M)	0,6	
Constant of Frequency (CF)	30	30
Force Multiplier (Fo_M)	1	1
Posture Multiplier (Po_M)	0,6	0,7
Additional Multiplier (Ad_M)	1	1
Repetitiveness Multiplier (Re_M)	0,7	1
Reference number of technical actions without Rc_M	5 796	9 660
Reference number of technical actions (RTA)	3 477,6	5 796
Cycle time	**30,0**	**30**
Total number of Actual Technical Actions observed (ATA)	19 320	7 360
Frequency (number of technical actions/min)	42,0	16,0
Number of technical actions in the cycle	21	8
Duration Multiplier (Du_M)	1	
***OCRA* index**	**5,6**	**1,3**

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Table H.6 — OCRA index in task B (net duration of the repetitive task equal to 460 min)

	right	left
Shift duration	480	
Breaks (min)	20	
Non repetitive work time (min)	0	
Work time considered as recovery (min)	0	
	460	
Write X for each task analysed	x	x
Repetitive work net time	460	
Number of cycles per shift	**460**	
Number of hours without an adequate	4	
recovery period (h)	0	
Recovery Multiplier (Rc_M)	0,6	
Constant of Frequency (CF)	30	30
Force Multiplier (Fo_M)	1	1
Posture Multiplier (Po_M)	0,7	0,7
Additional Multiplier (Ad_M)	1	1
Repetitiveness Multiplier (Re_M)	1	1
Reference number of technical actions without Rc_M	9 660	9 660
Reference number of technical actions (RTA)	5 796	5 796
Cycle time	**60,0**	**60,0**
Total number of Actual Technical Actions observed (ATA)	14 720	7 360
Frequency (number of technical actions/min)	32,0	16,0
Number of technical actions in the cycle	32	16
Duration Multiplier (Du_M)	1	
***OCRA* index**	**2,5**	**1,3**

H.3 Multi-tasks analysis

H.3.1 Computation of the overall number of Actual Technical Actions (ATA) in task A and task B (Table H.7)

In 460 min (480 min less 20 min of recovery period), one worker can rotate between task A (200 min) and task B (260 min). The other worker rotates in a reciprocal way.

Maintaining the same cycle times in each task, in 200 min (task A) the production will be 400 units, in 260 min (task B) the production will be 260 units.

The overall number of Actual Technical Actions needed (ATA) to perform the different repetitive tasks, is given by the following equation:

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$$ATA = \sum_{J=1}^{n} \left(FF_j \times D_j \right)$$

where

D_j is the foreseeable net Duration (in minutes) of the task j.

FF_j is the foreseeable frequency of actions per minute of task i;

In the present example:

ATA (right upper limb) = task A (42 × 200) + task B (32 × 260) = 16 720

ATA (left upper limb) = task A (16 × 200) + task B (16 × 260) = 7 360

H.3.2 Computation of the overall number of reference technical actions (RTA) in task A and task B (Table H.7)

The following general equation calculates the overall number of reference technical actions within a shift (RTA):

$$RTA = \sum_{J=\cdots 1}^{n} \left[CF \times \left(Fo_{Mj} \times Po_{Mj} \times Re_{Mj} \times Ad_{Mj} \right) \times D_j \right] \times \left(Rc_M \times Du_M \right)$$

In the present example:

task A (right upper limb) $\left[CF \times \left(FO \times Po \times Re \times Ad \right) \times D \right]$

[30 × (1 × 0,6 × 0,7 × 1) × 200] = 2 520

task B (right upper limb) $\left[CF \times \left(FO \times Po \times Re \times Ad \right) \times D \right]$

[30 × (1 × 0,7 × 1 × 1) × 260] = 5 460

[task A (right upper limb)+ task B (right upper limb)] = 2 520 + 5 460 = 7 980

RTA (right upper limb) = 7 980 × ($Rc \times Du$) =4 788

RTA (right upper limb) = 7 980 × (0,6 × 1) = 4 788

task A (left upper limb) $\left[CF \times \left(FO \times Po \times Re \times Ad \right) \times D \right]$

[30 × (1 × 0,7 × 1 × 1) × 200] = 4 200

task B (left upper limb) $\left[CF \times \left(FO \times Po \times Re \times Ad \right) \times D \right]$

[30 × (1 × 0,7 × 1 × 1) × 260] = 5 460

[task A (left upper limb)+ task B (left upper limb)] = 4 200 + 5 460 = 9 660

RTA (left upper limb) = 9 660 × ($Rc \times Du$) = 5 796

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RTA (left upper limb) = 9 660 × (0,6 × 1) = 5 796

where

CF	is the “constant of frequency” of technical actions per minute = 30;
Fo_M ,Po_M, Re_M, Ad_M	are multipliers for the risk factors postures, repetitiveness, additional, force in the repetitive tasks A and B;
D	is the foreseeable net duration (in minutes) of the repetitive task A = 200 min and task B = 260 min;
Rc_M	is the multiplier for the risk factor “lack of recovery period” = 0,6 corresponding at 2 breaks of 10 min each in 460 min plus lunch break (see Annex F);
Du_M	is the multiplier according to the overall duration of all repetitive tasks during a shift = 1.

H.3.3 Computation of the overall number of reference technical actions within a shift in task A and task B (Table H.7)

$$OCRA = \frac{ATA}{RTA}$$

In the present example:

$OCRA$ index (task A and B right upper limb) = 16 720/4 788 = 3,5 (yellow zone)

$OCRA$ index ($task$ A and B left upper limb) = 7 360/5 796 = 1,3 (green zone)

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Table H.7 — The *OCRA* multi-task index for task A and B

Shift duration	480			
Breaks (min)	20			
Non repetitive work time (min)	0			
Work time considered as recovery (min)	0			
	A	B		
Insert "X" for each analysed task	x	x		
Repetitive work net time	200	260		
No. of cycles per shift	**400**	**260**		
Number of hours without an adequate	4	4		
Recovery period (h)	0	0		
Recovery Multiplier (Rc_M)	0,6	0,6		
Constant of Frequency (CF)	30	30		
	right		left	
	A	B	A	B
Force Multiplier (Fo_M)	1	1	1	1
Posture Multiplier (Po_M)	0,6	0,70	0,7	0,70
Additional Multiplier (Ad_M)	1	1,00	1,00	1,00
Repetitiveness Multiplier (Re_M)	0,7	1,00	1,00	1,00
Reference number of technical actions without Rc_M	2 520	5 460	4 200	5 460
Reference number of technical actions (RTA)	4 788		5 796	
Cycle time	**30**	**60**	**30**	**60**
Total number of Actual Technical Actions observed (ATA)	8 400	8 320	3 200	4 160
Frequency (number of technical actions per minute)	12,0	32,0	16,0	16
Number of technical actions in the cycle	21	32	8	16
Duration Multiplier (Du_M)	1	1	1	1
Ocra index	**2,5**		**1,3**	

Key

A Work place No. 1

B Work place No. 2

H.4 Conclusion

In this example one operator works at task A with *OCRA* index in red zone, the second works at task B with OCRA index in green zone: applying a turn over (each operator makes the two tasks A and B) the exposition will be in yellow zone for the right upper limb and in green zone for the left.

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